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1. Flight instructor refresher seminar
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2. Contents and requirements of training course
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61.01.5 MAINTENANCE OF COMPETENCY AND SKILLS TESTS

1. Revalidation Check

The contents of the revalidation check shall be as defined below and shall be conducted in an aircraft of the category, class or type for which the pilot requires the revalidation check or in a flight simulation training device (FSTD) approved for the purpose. In general terms, the elements listed below should be included in the revalidation check, however, it is accepted that there may be aircraft or operational requirements and limitations that prevents some of these elements from being covered. In such cases the DFE shall make appropriate comments on the revalidation check form. Where possible, in the interests of safety and for the purposes of conducting a realistic revalidation check, an FSTD should be used. The Director shall have final authority as to whether the items not covered constitute an acceptable revalidation check.

1.1 Normal Procedures

(1) Mass and balance data
(2) Take-off and landing distance or landing area requirements
(3) Altitude capability/flight planning
(4) Weather interpretation
(5) Filing of flight plan
(6) Pre-flight inspection
(7) Pre-start checks
(8) Starting and after start procedures
(9) Taxiing/hover checks
(10) Pre-take-off procedures and checks
(11) Crew/pilot briefing
(12) Departure and after departure procedures
(13) Climb procedures including best rate/maximum angle and cruise climb techniques and engine monitoring procedures
(14) Cruise techniques
(15) Use of navigation systems
(16) Use and monitoring of automation
(17) Descent techniques
(18) Approach preparation and briefing
(19) Flying the published instrument approach and relevant procedures
(20) Landing/hover techniques
(21) After-landing procedures
(22) Shutdown procedures
(23) Paperwork requirements including technical logs

1.2. Non-normal Procedures

Non-normal procedures require that the pilot, whilst controlling the aircraft, identifies and assesses the problem, carries out the appropriate action to contain the problem or malfunction and then applies appropriate management techniques to minimise the danger to the crew and passengers. The use of the correct quick reference handbook (QRH) and/or other appropriate flight documentation is an essential element of successfully completing the proficiency or re-validation check.

Non-normal events and malfunctions include but are not limited to the following:

(a) Operation on wet or contaminated runways
(b) Operation in strong crosswinds
(c) Wind-shear recovery procedures
(d) CFIT recovery procedures
(e) Aborted and alternate engine start procedures
(f) Rejected take-off
(g) Engine failure procedures
(h) Autorotation
(i) Vortex ring recovery
(j) Mast bumping
(k) System failure procedures
(l) Instrument failure procedures
(m) Avionic and auto-flight control failure procedures
(n) Radio failure procedures

1.3. Crew Procedures

(1) Crew procedures include but are not limited to the following:
- (a) CRM
- (b) Threat and error identification and management
- (c) Multi-crew co-operation
- (d) Communication including ATC communications
- (e) General management of the flight
- (f) Situational awareness
- (g) Declaring an emergency
- (h) Pilot incapacitation procedures in multi-crew aircraft

(2) In the case where the applicant holds an instrument rating the revalidation check should be carried out under IFR and in IMC or simulated IMC as far as possible. The applicant should act as pilot flying unless the requirement for dealing with a non-normal procedure or the operator's procedures requires otherwise.

1.4. Revalidation Check Form

The revalidation check shall be conducted using the applicable form(s) as published on the SACAA website.

2. Summary

The summary shall:

(1) Reflect: the number of hours flown recorded in each column of the logbook, per aircraft category, class or type in the preceding 12 months, as well as a grand total for the period; and

(2) Be submitted together with annual currency fees and attached to the form required for the applicable revalidation or competenc check.

61.01.7 LANGUAGE

1. Proficiency requirements

(1) ICAO has specified English Language Proficiency Requirements, and mandated that these requirements shall be effective from 5 March 2008. There are six levels of proficiency in the requirement (refer to Appendix 1 of ICAO Annex 1 and the ICAO Language Proficiency Rating Scale attached to ICAO Annex 1).

(2). In accordance with the requirements, Pilots and Air Traffic Service Personnel shall demonstrate a minimum proficiency of at least Operational Level '4' of both ICAO Standard Phraseology and plain language, to be issued with or to maintain their respective licences.

(3). The requirements further indicate that Pilots and Air Traffic Services Personnel who have not been rated at Level 6 proficiency shall be tested for English language proficiency at regular intervals to ensure that they remain proficient at the required level. Pilots and Air Traffic Service Personnel who have been rated at Level 6 proficiency shall not require retesting.

(4). The six language proficiency levels are:

PROFICIENCY LEVEL	PROFICIENCY TESTING INTERVAL
Level 6: Expert	Retesting not required
Level 5: Extended	Retesting required every six years
Level 4: Operational (Minimum level)	Retesting required every three years
Level 3: Pre-operational	Licence not issued/maintained
Level 2: Elementary	Licence not issued/maintained

Level 1: Pre-elementary	Licence not issued/maintained

(5). Language Proficiency Requirement applies to speaking and listening proficiency only and does not address the ability to read or write, in the English Language.

(6). Areas of language to be assessed: The following six dimensions as prescribed by ICAO Annex 1 are listed as:

(a) Pronunciation: Ability to speak in a manner that is clear and easy to understand.

(b) Structure: Ability to compose concise, meaningful and unambiguous sentences or messages.

(c) Vocabulary: Ability to use correct words and phrases to match the setting.

(d) Fluency: Ability to respond, narrate events or describe situations naturally.

(e) Comprehension: Ability to understand and follow instructions without difficulty.

(f) Interaction: Ability to ask and answer questions, and engages in two-way dialogue without difficulty.

(7). Licences affected by ICAO language requirements: The following licences are affected by this requirement:

(a) Aeroplane and helicopter pilots.

(b) Glider and free balloons.

(c) Air Traffic Service Personnel.

(d) Aeronautical station operators.

(e) Exempted from the requirement:

(i) Aircraft maintenance engineers.

(ii) Flight dispatchers.

(8). Compliance with language proficiency procedures:

(a) In order to comply with this requirement all Flight crew and Air traffic Service Personnel shall be tested and rated accordingly.

(b) Approved Language Proficiency Interviewers/Raters will be registered with the CAA and shall comply with ICAO language proficiency requirements as set out in ICAO Document 9835, Ch 5 & 6. Language Proficiency Ratings shall comply with the critical characteristics of testing methodology as set out in ICAO Document 9835, Ch 6 (6.6.3) and make use of the ICAO Rating Scale, (See Appendix 1.5.1 to SA-CATS 61) and holistic descriptors provided.

(c) Approved Training Organizations shall apply to the CAA for accreditation, whereby individual instructors at these organizations are registered as Language Proficiency Interviewers/Raters in accordance with their respective professional and academic qualifications. (See ICAO document 9835, Ch 4 (4.3.1) and Appendix 1.5.2 to SA-CATS 61 for best practices.)

(d) Language Proficiency Interviewing/Rating is considered to be extremely “high stakes” testing as it impacts on aviation careers and the critical issue of aviation safety. To ensure validity of such assessments, it is imperative that the interviews and ratings shall only be conducted by professionals registered as CAA Language Proficiency Interviewers/Raters.

(e) Oral Proficiency Interviews (OPI’s) must be conducted jointly by a CAA registered Subject Matter Expert (SME) and a Linguistic Expert. All OPI’s shall be conducted using face-to-face interview procedures, and each OPI shall be recorded (preferably electronically) and stored for a minimum of 6 years in an archive system. These records shall be open for inspection and audits by the CAA.

(f) All CAA registered Language Proficiency Interviewers/Raters shall demonstrate full proficiency (Level 6 competency) in their language usage. They shall also be required to sign a Code of Conduct concerning language testing practices.

(g) Approved Test Centres and Language Proficiency Interviewers/Raters shall hold SAQA accreditation.

(9). The pre-interview process:

The pre-interview process comprises completion of a Bio-Data Questionnaire included under Part 1 on the CAA Language Proficiency Test Report for Radio Telephony Communications (Form CA 61-01-07).

(10). The interview process, certification and endorsement:

(a) The interview process shall consist of a thirty to forty minute interview with a Linguistic and a Subject Matter Expert (pilot or ATC). During this interview, the candidate shall be evaluated in the use of the plain language and aviation related language. Oral and listening skills shall be rated in accordance with the full set of ICAO descriptors.

(b) This process shall determine the candidate's actual proficiency level. The candidate shall then be issued with a Certificate of Language Proficiency from the Test Centre that will be submitted together with Form CA 61-01-11 for issuing of a rating into the candidate's licence. The applicable fee for the issuance of a rating as listed in CAR Part 187 is payable to the CAA.

(c) The Test Centre shall determine beforehand the test category that is most appropriate for the candidate, based on the candidate's profession. Separate tests shall be used for Pilots and Air Traffic Service Personnel who are considered ab initio or professional as per the following:

(i) The holder of a Commercial Pilot Licence shall be considered a professional pilot, whereas a pilot who flies in their private/personal capacity shall be considered to be *ab initio.*

(ii) Air Traffic Service personnel who hold a validated ATS rating shall be considered professional. All other Air Traffic Personnel shall be considered as ab initio.

(11) The certification process:

Approved Test Centres shall issue an English Language Proficiency Certificate that shall include the following minimum criteria:

(a) Name of the certificate, i.e. Certificate of Competency in ICAO English Language Proficiency.
(b) Name of the Test Centre.
(c) CAA approval number for the Test Centre.
(d) Identity Number of the applicant.
(e) Licence number of the applicant.
(f) Profession of the applicant.
(g) Full names of the applicant.
(h) Colour identity photograph of the applicant.

(i) The assessed rating result in all six skill dimensions reported under Part 2 Section 1 of Form CA 61-01-11, i.e.:

(i) Pronunciation.
(ii) Structure.
(iii) Vocabulary.
(iv) Fluency.
(v) Comprehension.
(vi) Interaction.

(j) Overall ICAO Language Proficiency Rating, reported under Part 2 Section 1 of Form CA 61-01-7.
(k) Name, Signature and CAA approval number of the Linguistic Expert.
(l) Name, Signature and CAA approval number of the Subject Matter Expert.

(12) Acceptance of Prior Learning or Foreign CAA language certification:
(a) The SA CAA will accept foreign CAA language certification and issue the relevant rating into the applicant's licence.
(b) The process for alternative language certification is:
(i) The CAA will certify the holder of a South African Private Pilot Licence or Commercial Pilot Licence at Operational Level 4.
(ii) The CAA will certify the holder of an Instrument Rating for a South African Pilot licence at Extended Level 5.
(iii) An applicant who can provide evidence of Expert English Language Proficiency will be awarded Expert Level 6. This evidence is indicated, but not limited to the following options:
(aa). Certified copy of either a school leavers certificate or a statement of results indicating Matric, O or M Level, with a pass in English first language with a minimum symbol of D or its equivalent.
(bb) Certified copy of a SAQA recognised 2 year tertiary qualification with English either as a subject or English as the language of tuition and examination.
(cc) Proof of being a present or past native of a nation where English is the first language, e.g. UK, USA, Australia or New Zealand.
(iv) An applicant who can provide evidence of Operational English Language Proficiency will be awarded a Level 4. This evidence is indicated, but not limited to the following options either a school leaver's certificate or a statement of results indicating Matric, O or M Level, with a pass in English second language with a minimum symbol of C or its equivalent.
(c) Candidates applying for an English rating proficiency in terms of (b)(ii) or (b)(iii) above are required to submit a declaration that he or she does not suffer from a significant speech impediment or have a heavy accent that would make the spoken English language difficult to understand.
(13) Candidates assessed to be below Level 4:

(a) Candidates who do not meet the ICAO Level 4 requirement, will be required to undergo remedial language training, in order to address areas of difficulty identified in Part 3 of Form 61-01-7; – Language proficiency feedback.
(b) Applicants assessed below Level 4 as well as applicants assessed at Level 4 and level 5 who wish to gain a Level rating shall be required to wait for a period of 90 days after being assessed before applying for re-assessment.
(c) A moderation process should be established by the Test Centres should candidates wish to challenge their results.

14. The linguistic and subject maater expert requirements can be found in the applicable appendix to this Part.

61.01.8 LOGGING OF FLIGHT TIME

1. Pilot-in-command-under-supervision (PICUS) functions

(1) The functions that the pilot-in-command-under-supervision are required to carry out, without intervention by the supervising designated pilot-in-command include, but are not limited to –

(a) Checking the accuracy of the proposed flight plan and the load sheet for the flight, including the computation of fuel;

(b) Ensuring that all checks are carried out in accordance with the check system established by the operator;

(c) Ensuring in any emergency that the procedures contained in the operations manual or other relevant documents have been complied with by each crew member;

(d) Carrying out all the duties and functions of a pilot-in-command;

(e) Taking all decisions relating to the use of any flight and ground systems required in the case of operations conducted by automatic means;

(f) Ensuring that all problems occasioned by meteorological conditions, communications and air traffic control procedures are resolved; and

(g) The operator shall have an approved scheme in terms of which the pilot-in-command under supervision training is conducted.

(2) The supervising designated pilot-in-command shall certify an appropriate entry in the pilot-in-command under supervision's logbook.

2. Crediting of flight time for a flight examiner

A flight examiner may be credited for flight time as indicated in Document SA-CATS 61.36.

61.01.9 CREDITING OF FLIGHT TIME AND THEORETICAL KNOWLEDGE

1. Recognition of prior learning and experience by SAAF pilots and navigators

(1) Credit at ATPL level is only applicable to Pilot Instructor, Navigator Instructors applicants or SA Air Force Graduate Test Pilots of one of the following Society of Experimental Test Pilots (SETP) recognised test pilot schools.

(a) Empire Test Pilot School. (UK)

(b) United States Air Force Test Pilot School. (USA)

(c) United States Navy Test Pilot School. (USA)

(d) EPNER. (France)

(e) National Test Pilot School. (USA)

(f) International Test Pilot School. (UK)

(2) SA Air Force pilots and navigators may be exempted from all or some of the requirement to attend a ground school for the different licences or ratings as the case may be, issued under this Part, but are required to write the examinations reflected below except where credit is given for prior learning.

(3) The exemption may be granted to applicants who hold, or have held within the 60 months preceding the date of application, an active South African Air Force Pilot or Navigator qualification

(4) An exemption may also be granted in the case of an applicant who previously held a South African Air Force (SAAF) Pilot or Navigator qualification and obtained, and maintained competency of, a SACAA private pilot licence or higher issued under this part within 60 months of terminating active flying status in the Air Force provided that application is made no longer than 24 months after obtaining the SACAA licence.

(5) The application for exemption shall be accompanied by –

(a) a Curriculum Vitae describing the applicant's SA Air Force Career and details of their flying experience;

(b) A letter from the Officer Commanding of the squadron or unit where the applicant has served or is serving, confirming current active flying status or, where applicable, the date of last active flying

status. This is necessary to confirm that the applicant complies with the 12 month requirement as stipulated above;

(c) Confirmation of hours flown and types of aircraft flown in the SAAF (Certified copy of logbook must be signed out by Officer Commanding, CFI or responsible person;

(d) Certified copy of the applicant's ID document;

(e) Proof of instrument rating (if applicable);

(f) Certified copy of any SA CAA licence held;

(g) An explanation of the hours flown as summarised in the logbook:

(i) 140 hrs dual on PC7 (single-engine turbine (A));

(ii) 400 hrs PIC on PC7 (single-engine turbine (A));

(iii) 800 hrs Instruction;

(iv) 25 hrs dual on DC3 (multi-engine piston (A));

(v) 250 hrs Co-pilot on DC3 (multi-engine piston (A));

(vi) 1300 hrs PIC on DC3 (multi-engine piston (A));

(6) Examinations

(a) A person who only has an aeroplane pilot qualification in the South African Air Force requiring to be issued with a Pilot Licence (Helicopter), shall write the technical examinations for Aircraft Technical and General (H) at PPL, CPL or ATPL level as applicable;

(b) A person who only has a helicopter pilot qualification in the South African Air Force requiring to be issued with a Pilot Licence (Aeroplane), shall write the technical examinations for Aircraft Technical and General (A) at PPL, CPL or ATPL level as applicable.

(7) Exemption at PPL level

(a) A person who only has an aeroplane pilot qualification in the South African Air Force requiring to be issued with a Pilot Licence (Helicopter), shall write the technical examinations for Aircraft Technical and General (H) at PPL, CPL or ATPL level as applicable;

(b) A person who only has a helicopter pilot qualification in the South African Air Force requiring to be issued with a Pilot Licence (Aeroplane), shall write the technical examinations for Aircraft Technical and General (A) at PPL, CPL or ATPL level as applicable.

(8) Exemption at CPL level.

(a) A person who has qualified as a pilot or navigator in the South African Air Force (Aeroplane) requiring to be issued with a Commercial Pilot Licence (Aeroplane) and had a minimum of 500 hours operational experience on SA Air Force multi-engine transport/marine type aeroplanes, or SA Air Force multi-engine helicopters, may be exempted from all technical examinations except Air Law or if an instrument rating is required as part of the licence, then Air Law and Procedures.

(b) A person who has qualified as a pilot or navigator in the South African Air Force (Aeroplane) requiring to be issued with a Commercial Pilot Licence (Aeroplane) and who does not have a minimum of operational experience on SA Air Force multi-engine transport/marine type aeroplanes, or SA Air Force multi-engine helicopters may be exempted from all technical examinations except Flight Performance and Planning and Air Law or if an instrument rating is required as part of the licence, then Air Law and Procedures;

(c) A person who has qualified as a pilot or navigator in the South African Air Force (Helicopter) requiring to be issued with a Commercial Pilot Licence (Helicopter) and had a minimum of 500 hours operational experience on SA Air Force multi-engine transport/marine type aeroplanes, or SA Air Force multi-engine helicopters, may be exempted from all technical examinations except Air Law or if an instrument rating is required as part of the licence, then Air Law and Procedures.

(d) A person who has qualified as a pilot or navigator in the South African Air Force (Helicopter) requiring to be issued with a Commercial Pilot Licence (Helicopter) and who does not have a minimum of 500 hours operational experience on SA Air Force multi-engine transport/marine type aeroplanes, or SA Air Force multi-engine helicopters may be exempted from all technical

examinations except Flight Performance and Planning and Air Law or if an instrument rating is required as part of the licence, then Air Law and Procedures.

(9) Exemption at ATPL level

(a) A person who has qualified as a pilot instructor or navigator instructor in the South African Air Force (Aeroplane) requiring to be issued with an Airline Transport Pilot Licence (Aeroplane) and had a minimum of 500 hours operational experience on SA Air Force multi-engine transport/ marine type aeroplanes, or SA Air Force multi-engine helicopters, may be exempted from all technical examinations except Air Law and Procedures.

(b) A person who has qualified as a pilot instructor or navigator instructor in the South African Air Force (Aeroplane) requiring to be issued with an Airline Transport Pilot Licence (Aeroplane) and who does not have a minimum of 500 hours operational experience on SA Air Force multi-engine transport/marine type aeroplanes, or SA Air Force multi-engine helicopters may be exempted from all technical examinations except Flight Performance and Planning as well as Air Law and Procedures.

(c) A person who has qualified as a pilot instructor or navigator instructor in the South African Air Force (Helicopter) requiring to be issued with an Airline Transport Pilot Licence (Helicopter) and had a minimum of 500 hours operational experience on SA Air Force multi-engine transport/ marine type aeroplanes, or SA Air Force multi-engine helicopters, may be exempted from all technical examinations except Air Law or if an instrument rating is required as part of the licence, then Air Law and Procedures.

(d) A person who has qualified as a pilot instructor or navigator instructor in the South African Air Force (Helicopter) requiring to be issued with an Airline Transport Pilot Licence (Helicopter) and who does not have a minimum of 500 hours operational experience on SA Air Force multiengine transport/marine type aeroplanes, or SA Air Force multi-engine helicopters may be exempted from all technical examinations except Flight Performance and Planning and Air Law or if an instrument rating is required as part of the licence, then Air Law and Procedures.

(10) Exemption for a Flight Instructor Rating

The holder of a valid South African Air Force flight instructor, navigator instructor or test pilot qualification on individual application and assessment:

(a) Is exempted from all components of the written flight instructor examinations.

(b) Is exempted from all or some of the requirement to attend a ground school.

(c) Shall in the case of a South African Air Force flight instructor undergo the ground evaluation test and the skills test for the issue of a Grade III Flight Instructor Rating as required by CAR 61.12.1 (e), (f) and CAR 61.15.1 (e), (f).

(d) Shall in the case of a South African Air Force flight instructor conduct at least 20 hours of flight instruction as a Grade III flight instructor before being eligible to upgrade to either a Grade II or I flight instructor.

(e) Shall in the case of a South African Air Force navigator instructor or test pilot graduate conduct at least 20 hours of patter before being eligible to undergo the ground evaluation test and the skills test for the issue of a Grade III Flight Instructor Rating as required by CAR 61.13.1 (e), (f) and CAR 61.15.1 (e), (f).

(f) Shall in the case of a South African Air Force flight instructor rating held in only one category of aircraft and an exemption from the flight instructor technical examinations is requested in a different category, then write the flight instructor examination for Principles of Flight in the new category.

61.01.10 THEORETICAL KNOWLEDGE EXAMINATIONS

1. Examination procedure: CPL, IF and ATPL candidates

(1) General Information

(a) An applicant for a pilot licence, or an instrument rating, shall demonstrate a level of knowledge appropriate to the privileges of the licence or rating for which application is made by passing theoretical knowledge examinations in accordance with the procedures set out in respective licence SA-CATS 61 Appendix 1.0 (PPL) or SA-CATS 61 Appendix 2.0 (IR/CPL/ATPL).

(b) Questions appropriate to the syllabi will be held in a CAA Central Question Bank (CQB). The examinations will be provided in English, using abbreviations where applicable and compiled by a computer in multiple-choice format. A list of common abbreviations used in the examinations can be found in Part 1 of the Regulations.

(c) Each multiple choice question shall generally have three or four alternative choices. Marks will not be deducted for incorrect selection.

(2) Theoretical Knowledge Examination Bookings

(a) Candidates should apply by completing the appropriate application forms, available on the website www.caa.co.za or at an approved CAA examination centre. No bookings can be made by telephone, and all bookings are made on a first come, first served basis.

(b) Candidates should indicate on the application form their first and second preference venue choices (written exams) and the date they wish to sit the examinations. The examinations section will contact applicants where dates requested cannot be met and in the case of a venue being fully booked an alternative will be offered. The Director reserves the right to change any venue to satisfy demand and does not guarantee a candidate a specific venue or examination date.

(c) Examination fees are payable directly to the CAA accredited examination centres or in the case where examinations are undertaken at the offices of the CAA, fees must be submitted with the application form, where applicable. Consult the CAA website for details of payment methods. Bookings will not be confirmed unless all the requirements are met. Once a booking has been made, candidates will receive an examination booking confirmation.

(d) The time between the closing date for applications and the examination sittings is two weeks. The Director will endeavour to send booking confirmations, venue details and examination timetables to reach candidates within 2 days of the booking and, in any case, at least 1 week prior to the sitting.

(e) Details of the scheduled examination dates and venues can be found on the CAA website and in AIC 31.1.3.

(f) Due to the restriction on the number of sittings allowed, the candidates on initial application, shall be allowed to book in advance for the entire series of examinations in a sitting or sittings. The candidate shall also have the privilege of booking for all rewrites in subsequent sittings.

(3) Booking/Cancellation/Venue Transfer Procedure

(a) Examination bookings may not be changed in the week prior to the examination. Cancellations will only be accepted, if received in writing or on-line, at least five working days before the examination. For CAA purposes five working days means Monday to Friday (excluding public holidays). Only under exceptional circumstances and upon written application, will the Director consider refunding examination fees;

(b) Under exceptional circumstances and upon written application, the Director may allow the candidate to transfer to an alternate venue.

(4) Attendance at the Examination

(a) Candidates should be present at the examination centre at least 30 minutes before the scheduled time for the commencement of each examination paper, with the following documents –

(i) An acceptable form of identification (Identity book, Driver's licence or Flight Crew Licence or Passport).

(ii) Confirmation of booking.

(iii) In the case of an initial writing of a subject, proof of having undergone the required knowledge instruction signed by the Chief Ground Instructor or his designee of a Part 141 approved training organisation.

(iv) Candidates may only enter the examination room during the 10 minutes preceding the start of the examinations to prepare examination material. They must not remain in the room after the finish of the examination period.

(v) On occasions when there is a possibility of disruption to transport services, for whatever reason, candidates are expected to make alternative arrangements for attendance or, if appropriate, to give formal notice of their inability to attend. Personal coats, bags, briefcases, etc. may be placed at the front/rear of the examination room, under the direction of the invigilating officer. Any bags etc. could be removed if left unattended outside the examination room.

(vi) Note: The Director accepts no responsibility for items of personal equipment a candidate brings into the examination hall and which he/she is not permitted to retain during the examination. Whilst every attempt is made to ensure reasonable comfort in examination halls, the Director cannot be held responsible for extraneous noise or for any breakdown or fluctuation in heating, lighting or ventilation facilities in examination halls which are operated on hire or lease arrangements and over which the Director, as a result, has no direct control. Candidates are also advised that, at all examination centres, a 'no smoking rule' must be observed.

(vii) Candidates must understand that failure to comply with these requirements will bar them from writing that particular examination.

(5) Materials for the Examination

(a) When necessary the required reference books and tables will be supplied to each candidate but they must not be marked in any way or removed from the examination room.

(b) Candidates are required to provide themselves with all the necessary drawing and calculating instruments, e.g. dividers, compasses, protractors, parallel rules, slide rules, navigational computers and a scientific, non-programmable, non-alphanumeric. Candidates may use their own pens, pencils, highlighters etc. on the rough working paper provided and/or on their own documents. Documents provided by the Exam Centre must not be marked in any way. No pencil boxes, containers or instrument cases are permitted on tables.

(c) The use of slide rules or instruments containing printed information on critical point, point of no return, distance to the horizon, convergence, conversion angle, departure, dlat, dlong, conversion factors, etc. is not permitted.

(6) Examination Briefing

(a) Before the start of the examinations, the invigilator will give a briefing regarding the examination.

(b) Regulations which will be applied to the conduct of examinations:

(i) Candidates are not allowed to use any loose paper other than that provided at the examination. All papers issued and documents provided by the Exam Centre are to be returned with the answer sheet to the invigilator on completion.

(ii) Answer sheets must be completed in ink, where applicable. Candidates may use other writing implements on the rough working paper.

(iii) Candidates must ensure that, where required, all answers have been transferred onto their answer sheet by the end of the examination. Candidates failing to do this will not be given any extra time.

(iv) Silence is to be observed in the examination room at all times. Electronic alarms and key rings are not permitted. Mobile telephones, pagers etc. must be switched off and left in the candidates' personal belongings.

(v) If a candidate wishes to speak to an Invigilating Officer, he/she should remain seated and raise his/her hand. It should be noted that the Invigilating Officer will consider only those questions from candidates that relate to the general conduct of the examinations and he/she will not enter into discussion on the interpretation of words or questions contained in the examination papers.

(vi) A candidate may leave the room with the permission of the invigilating officer if he/she finishes an examination before time, except during the last 5 minutes before the end of any paper. Candidates are to stop work and put pencils or pens down when so directed and must remain seated and quiet until all answer material has been collected.

(vii) Any candidate who attempts to remove unauthorised examination materials from the room will be liable to disqualification from those examinations that have been taken and may be subject to special arrangements for future examinations.

(viii) Refer to AIC 31.1.3 for additional examination rules.

(7) Failure to comply with examination regulations

Any infringement of examination regulations may result in the candidate being disqualified in any subject that has been taken and may be barred from further participation in future examinations at the discretion of the Director.

(8) Results

(a) Results will normally be available immediately at on-line Examination Centres or will be dispatched by mail within 10 working days following the end of the examination week concerned. Results will also be posted on the CAA website. Candidates should not telephone Examinations section to request examination results, as results will not be given over the telephone under any circumstances. In the event of non-receipt of a result notification, arrangements can be made for repeat notifications to be sent by post.

(b) The Director cannot enter into discussion or correspondence with candidates on the subject of their examination results, but candidates may apply for any paper to be remarked in terms of the CAR on payment of the fee prescribed in Part 187.

(9) Re-mark after failure

(a) A candidate, who fails an examination conducted by the Civil Aviation Authority, may, within 30 days from the date of notification of the examination results, apply in writing for a re-mark. Candidates who fail with a mark of 65% or above may request a remark.

(b) The application shall be accompanied by the appropriate fee prescribed in Part 187.

(c) If the re-mark is successful, the fee will be refunded.

(10) Re-write after failure

(a) A candidate, who fails an examination conducted by the Civil Aviation Authority, may apply in writing for a re-mark up to 30 days after the date of notification of the examination results.

(b) Candidates cannot apply to re-write examinations, which they believe they may have failed, until they have received the official result notification. Furthermore, applicants who have applied for a re-write may not apply for the examination being remarked until they receive the official examination result notification.

(c) Candidates for re-writes may not be allowed to re-write an examination, unless special circumstances prevail, within a period of 72 hours of an unsuccessful attempt.

(d) Candidates for re-writes, who achieved a mark of less than 65%, will be required to produce proof that they have completed additional knowledge instruction from a Part 141 approved training provider, at the time of the re-write.

(11) Theoretical Knowledge Examination Pass Standards

(a) A candidate must complete all required written CPL/ATPL/IR examination papers within 18 months of achieving a first credit for an examination. A credit for an examination is held for each successful attempt.

(b) The papers can be attempted in any order. A Pass in an examination paper will be awarded to a candidate achieving at least 75% of the marks allocated to that paper.

(12) Failure to comply with Pass Standards

(a) An applicant who does not pass all of the required PPL, CPL, Instrument Rating, ATPL or Flight Instructor theoretical knowledge examinations within a continuous period of 18 months will be required to re-write all theoretical knowledge examinations as though for an initial attempt.

(b) In the event of an applicant who is the holder of student pilot licence or a private pilot licence on an integrated course, then the 18 month time limit for credits towards an ATPL is calculated from a first time pass in the subjects listed in section 1(3)(a) of TS 61.08.1.

(13) Theoretical Knowledge Examination Credits

(a) Details of SA CAA Theoretical Examination Credits can be found in Table 2 of this Technical Standard.

(b) Candidates who obtain credit or a pass for the ATPL subjects have 36 months to obtain an Instrument Flight Rating. The ATPL subjects will remain valid for a period of 60 months from the date of expiry of the last Instrument Flying Revalidation Check.

(c) Candidates who obtained a pass at ATPL level under the provisions of the ANR's of 1976 and who have maintained an Instrument Flying Rating are afforded the same privilege as detailed in (b) above.

(14) Failure to obtain CPL/IR/Flight Instructor Rating within 36 month period

If a CPL and IR are not issued within the 36 month period from the date of passing the last CPL/IR or ATP examination as the case may be, then the CPL/IR and ATPL theory credit for

Air Law and Procedures will lapse. Candidates will be required to re-take the Air Law and Procedures CPL/IR knowledge examinations in order to regain CPL/IR or ATPL theory credits. Where a candidate has previously passed all ATPL theoretical knowledge examinations but was not issued with a CPL/IR within the 36 month period, the amount of credit to be given for the ATPL theoretical knowledge instruction for the Air Law and Procedures will be at the discretion of the Head of Training of the Part 141 approved training organisation. If the Flight Instructor Rating is not issued within 36 months of the date of the last Flight Instructor Rating examination, then the candidate will be required to re-take all the Flight Instructor Rating examinations.

(15) Credit of ATPL Examination

(a) In some circumstances a candidate who has previously attempted some ATPL theoretical knowledge examinations may wish to consider attempting examinations at a lower level (i.e. CPL and/or IR). A candidate who has failed to obtain a pass in any subject at ATPL level will be required to enter for the CPL and/or IR examinations as though for an initial sitting.

(b) A candidate who has previously completed an approved ATPL theoretical knowledge course may be credited with the CPL and/or IR theoretical knowledge course.

(c) Candidates will be required to enter for these theoretical knowledge examinations via an approved Part 141 Aviation Training Organisation. A candidate who has passed at least one subject at ATPL level may be credited the equivalent subject at CPL and/or IR level as detailed below.

(d) Candidates should note that where credit is given in accordance with Table 2, all sittings, attempts and time limits will be calculated from the initial attempt at the ATPL examinations.

Table 2: ATPL Examinations Credits in respect of CPL and IR

ATPL	CPL/IR	IR	CPL (VFR)	PPL
				
Aviation Meteorology	Aviation Meteorology	Aviation Meteorology	Aviation Meteorology	Aviation Meteorology
Flight Performance and Planning	Flight Performance and Planning	Flight Performance and Planning	Flight Performance and Planning	Flight Performance and Planning
Radio Aids and Communication	Radio Aids and Communication	Radio Aids and Communication	Radio Aids and Communication	
General Navigation	General Navigation	General Navigation	General Navigation	General Navigation
Navigation (Plotting)				
Instruments and Electronics	Instruments and Electronics	Instruments and Electronics	Instruments and Electronics	
Aircraft (A or H) Technical and	Aircraft (A or H) Technical and		Aircraft (A or H) Technical and	Aircraft (A or H) General

General	General		General	
				Principles of Flight (A or H)
Human Performance and Limitations				Human Performance and Limitations
*Air Law and Operational Procedures			*Air Law	Air Law

- *Note 1: A candidate who has decided to discontinue writing the ATPL examinations and who has a passed a subject at ATPL level will be credited with the subject at CPL/IR, IR or CPL (VFR). Should the candidate wish to upgrade his/her licence then all subjects will have to be rewritten at the correct level with the exception of Human Performance and Limitations and Air Law and Operational Procedures.*
- *Note 2: A candidate who has decided to discontinue writing the ATPL examinations and who has passed Human Performance and Limitations examination papers at CPL (VFR) level will be credited with the Human Performance and Limitations examination required at CPL/IR, IR or ATPL.*
- *Note 3: A candidate who has passed Air Law and Operational Procedures examination papers at ATPL level will be credited with the Air Law and Operational Procedures examination required at CPL/IR or IR.*
- *Note 4: The IR exams under Part 61 are only offered at the level of ATPL or CPL. A PPL requiring an instrument rating shall write the examinations at CPL level. These examinations will act as a credit for the CPL (IR) examinations provided that the applicant successfully attains an instrument flying rating within 36 months of attaining a credit for the examinations. The credit for examinations remain valid for a period of 60 months from the date of the last successful IR revalidation check.*
- *Note 5: The credits for the CPL (IR) examinations with the exception of Air Law/Operational Procedure and Human Performance are not applicable to the ATPL examinations.*
- *Note 6: The credits obtained for ATPL examinations are valid for the CPL (IR), CPL (VFR) and the IR examinations.*

(16) Crediting of Theoretical Knowledge

(a) Where an applicant holds the theoretical knowledge credits for an aeroplane licence and wishes to obtain an equivalent helicopter licence and vice versa then the following rules apply to the transfer of the credits held.

(b) In order to satisfy the theoretical knowledge requirements for the ATPL (H), the holder of an ATPL (A), or CPL (A) with valid ATPL (A) theory credit, is required to complete approved bridge instruction for the subjects/topics detailed in Appendix 1 to SA-CATS 61.8. In addition, a pass in the following ATPL (H) examinations must be obtained –

(i) Aircraft (H) Technical and General;

(c) In order to satisfy the theoretical knowledge requirements for the CPL (A), the holder of an ATPL (H) or CPL (H) is required to complete approved instruction for the subjects/topics detailed in Appendix 1 to SA-CATS 61.5. In addition, a pass in the following CPL (A) examinations must be obtained,

(i) Aircraft (A) Technical and General;

(d) Division 5 - Examination arrangements shall apply.
The Director shall publish dates, times and venues for the examinations in an AIC.

(e) ATPL Papers

(i) Aviation Meteorology;
(ii) Flight Performance and Planning;
(iii) Radio Aids and Communication;
(iv) General Navigation;

(v) Navigation (Plotting)
(vi) Instruments and Electronics;
(vii) Aircraft (A or H) Technical and General;
(viii) Human Performance and Limitations;
(ix) Air Law and Operational Procedures.

(f) CPL/IR Papers
(i) Aviation Meteorology;
(ii) Flight Performance and Planning;
(iii) Radio Aids and Communication;
(iv) General Navigation;
(v) Instruments and Electronics;
(vi) Aircraft (A or H) Technical and General;
(vii) Human Performance and Limitations;
(viii) Air Law and Operational Procedures.

(g) IR Papers
(i) Aviation Meteorology;
(ii) Flight Performance and Planning;
(iii) Radio Aids and Communication;
(iv) General Navigation;
(v) Instruments and Electronics;
(vi) Human Performance and Limitations;
(vii) Air Law and Operational Procedures.

(h) CPL (VFR) Papers
(i) Aviation Meteorology;
(ii) Flight Performance and Planning;
(iii) Radio Aids and Communication;
(iv) General Navigation;
(v) Instruments and Electronics;
(vi) Aircraft (A or H) Technical and General;
(vii) Human Performance and Limitations;
(viii) Air Law.

(i) PPL Papers
(i) Aviation Meteorology;
(ii) Flight Performance and Planning;
(iii) General Navigation;
(iv) Aircraft (A or H) General;
(v) Principles of Flight;
(vi) Human Performance and Limitations;
(vii) Air Law.

61.01.13 RECOGNITION, VALIDATION AND CONVERSION OF A FOREIGN PILOT LICENCE AND RATING

1. Application for validation of foreign pilot licence and ratings

(1) The PPL validation for VFR day operations may be issued by the CAA on confirmation of the validity of the licence. The PPL (VFR) validation will be endorsed with the following wording –
For private day VFR operations only, provided that – “The holder of this validation may only exercise the privileges of the licence if in compliance with CAR 61.01.13(5)(a)”.

(2) Prior to the privileges of the validation for the PPL (VFR) being exercised, ATO's are to ensure that the applicant first successfully completes;

(a) An appropriate South African Air Law examination at an approved ATO;

(b) A flight test at an approved ATO, with a Grade I or II flight instructor; and

(c) Completes a Density Altitude Tutorial conducted by at least a Grade III instructor;

(d) Has the above three items endorsed in his logbook by the testing instructors;

(e) Forwards proof to the CAA within 30 days of completion of the requirements in paragraphs (a), (b), (c) and (d) above.

61.01.15 TRAINING FOR ACQUIRING LICENSE , RATING OR VALIDATION

1. General

(1) All flying training shall be conducted in accordance with the requirements laid down for a licence and rating as stipulated in the various Subparts.

(2) The written approval by the Director to conduct flying training shall be kept by the person conducting the training and produced on demand to any authorised officer or inspector.

Flying instruction

(3) Flight instruction and authorisation to a student for solo flying shall be such as to ensure that an aircraft piloted by a student does not constitute a hazard to air navigation or endanger the safety of life or property.

Flight instruction syllabus

(4) Flying training shall be conducted in accordance with the flight instruction syllabus prescribed in the Subparts applicable to the licence or rating sought.

Local rules

(5) A flying training organisation shall prepare and keep accessible to all students a copy of its local rules, which shall be submitted to the Director for approval.

(6) In conjunction with the rules referred to in sub-paragraph (8) there shall be displayed a map clearly showing –

(a) the general flying area;

(b) the low-flying area;

(c) the simulated instrument flying area (if applicable); and

(d) the acrobatic and spinning area.

Experience requirement for the appointment of a chief flying instructor

(7) A chief flying instructor must be the holder of at least a Grade II flight instructor rating and must satisfy the Director that he/she has –

(a) flown at least 1000 hours total flying time; and

(b) given not less than 500 hours of flight instruction of which at least 200 hours shall be ab initio training.

Responsibilities of chief flying instructors

(8) A chief flying instructor shall be responsible for –

(a) ensuring that each dual and solo training flight is authorised by the holder of an appropriate and valid flight instructor rating or a person appointed by the Chief Flying Instructor for the specific flight or sequence required by these regulations;

(b) ensuring that all authorisations are properly entered in the appropriate flight authorisation book;

(c) ensuring that the flight authorisation book is correctly completed before each dual and solo flight. Such authorisation book shall at least show the following entries –

(i) Date of the flight;
(ii) Aircraft type and registration;
(iii) Name and licence number of pilot in command;
(iv) Name and licence number of student pilot;
(v) Estimated elapse time;
(vi) Actual time of take-off;
(vii) Actual time of landing;
(viii) Actual flying time;
(viii) Description of exercise/route;
(viii) Name and full signature of the authorising person;
(ix) Full signature of the student before the flight (OUT); and
(x) Full signature of the student after the flight (IN).

(d) ensuring that each solo training flight is personally supervised by the holder of an appropriate and valid flight instructor rating or a person appointed by the Chief Flying Instructor;

(e) the maintenance of flying discipline;

(f) ensuring that student pilots receive the following dual flight instruction –

(i) a dual flight check prior to each solo flight during his first 3 hours of solo flight. Such dual flight check shall be conducted in accordance with subparagraph (9); and
(ii) subsequently a minimum of 1 hour dual instruction for every 5 hours' flight time until a private pilot's licence is obtained;

(g) ensuring that a dual competency check is conducted before the student pilot is permitted to undertake his first solo flight;

(h) ensuring that before a student pilot is authorised to conduct his first solo flight (exercise 14), the instructor who conducted the solo competency check flight has endorsed the student's logbook in accordance with SA-CATS 61;

(i) ensuring that before a student pilot is authorised to leave the circuit area on a solo flight to the general flying area or on a solo navigation flight, a flight instructor has endorsed the student's logbook in accordance with SA-CATS 61;

(j) the correct maintenance of pilot's logbooks by pupils and student pilots who are under training at the organisation;

(k) ensuring that the aircraft is equipped in accordance with CAR 91;

(l) ensuring that the standard of ground instruction given shall comply with the standards required for the licence to be obtained;

(m) ensuring that a training record of each student or pilot trained is properly maintained. This record is the property of the student and shall company him or her as a portfolio of evidence. The training organisation is required to make a copy of the training record and is to keep such record in a safe place for a minimum of 60 months. However, the full training file and logbook of a student or pilot trained shall be kept in a secure area on the premises of the training organisation during the entire training period. The training record shall include:

(i) the full details of the student or pilot trained;

(ii) the name and licence number of the flight instructor concerned with the training;

(iii) a training progress report for each individual training session which shall include:

(aa) the name of the student or trained pilot;

(bb) the name of the instructor conducting the training session;

(cc) the description of the exercise/route;

(dd) detailed de-brief comments related to each training session;

(ee) records of the dual flying hours;

(ff) records of the solo hours conducted in the circuit and general flying area;

(gg) record of the solo hours conducted during navigation flights; and

(hh) a summary of the hours flown solo and dual;

(iv) a record of all the theoretical examinations;

(v) a record of all the briefings and courses attended pertaining to the licence sought;

(vi) certificates of competency;

(vii) all reports of passed and failed theoretical examinations;

(viii) an exercise checklist;

(ix) whether the training was successfully completed and the duration of the training period;

(n) ensuring that no student pilot will conduct a flight below a height of less than 500 feet unless accompanied by a holder of an appropriate and valid flight instructor rating.

(9) Dual progress check flights

(a) ensuring that a dual progress check flight is conducted at the latest after a student completed his/her first 15 hours dual flight instruction and after each completion of 10 hours flight time thereafter.

(b) the dual progress check flight must be conducted by the chief flying instructor or by an appointed Grade II or Grade I instructor.

(c) Each dual progress check flight must be endorsed by the checking instructor in the logbook of the student in accordance with SA-CATS 61.

61.01.17 APPROVAL OF FLIGHT SIMULATION TRAINING DEVICES

1. The approval process

The approval process shall be in accordance with Appendix C to Document SA-CATS 61.

61.02.1 REQUIREMENTS FOR STUDENT PILOT LICENCE

1. Theoretical knowledge course and examination

(1) The theoretical knowledge course and pre-solo theoretical examination shall cover the following aspects:

(a) Air Law, as appropriate to student pilots; and

(b) Aircraft Knowledge, covering the aircraft make and model used for training.

(2) The written theoretical knowledge examinations shall be conducted at an approved Aviation Training Organisation.

(a) The examination shall be conducted and corrected by the holder of an appropriately rated Grade I or Grade II flight instructor, Aeroplane or Helicopter, respectively;

(b) The flight instructor referred to in paragraph (a) may not be the flight instructor from whom the applicant received his or her theoretical training.

(3) The Communications Syllabus can be found in Appendix 1.5 to Document SA-CATS 61.

61.02.5 PRIVILEGES AND LIMITATIONS OF STUDENT PILOT LICENCE

Note: Entry to training

Before being accepted for training an applicant should be informed that the appropriate medical certificate must be obtained before solo flying is permitted.

1. Requirements for, authorisation and supervision of solo training flights

(1) A student cannot be released for the first solo flight unless he/she:

(a) has undergone a minimum of 10 hours of dual flight training;

(b) is holder of a student pilot licence;

(c) proves to possess adequate knowledge of the basic principles of flight;

(d) has undergone training in exercises 1 through 13;

(e) has shown proficiency in handling the aircraft in the event of an engine failure during initial climb-out and from downwind position;

(f) has shown proficiency in recovery from a balloon during landing and a bounced landing; and

(g) has shown proficiency in executing a go-around manoeuvre from a full flaps configuration.

(2) Each solo training flight must be authorised by the Chief Flying Instructor (CFI) or by the holder of a valid flight instructor rating appointed by the CFI for the specific flight or sequence required by these regulations.

(3) Each solo training flight authorisation must be properly entered in the appropriate flight authorisation book in compliance with the proper format represented in Part 141 of the Regulations.

(4) Each solo training flight must be personally supervised by the holder of a valid flight instructor rating or a person appointed by the Chief Flying Instructor.

2. Dual competency check flight, dual check flight and dual progress check flight

(1) A dual competency check flight must be conducted before the student pilot is permitted to undertake his first solo flight.

(2) The dual competency first solo check flight shall be conducted by the Chief Flying Instructor (CFI) or a Grade II or Grade I instructor appointed by the CFI.

(3) Before a student pilot is authorised to conduct his first solo flight (exercise 14), the instructor who conducted the dual competency check flight must endorse the student's logbook in accordance with SA-CATS 61.

(4) The dual competency first solo check flight must include but is not limited to –

(a) at least 3 take-offs and landings

(b) one glide approach to a landing

(c) one simulated engine failure during initial climb out

(d) one go-around from a full flaps configuration.

(5) A dual check flight shall be conducted by a suitably rated instructor prior to each solo flight during the first 3 hours of the student's solo flight time –

(a) subsequently a minimum of 1 hour dual instruction shall be conducted for every 5 hours solo flight time until a private pilot's licence is obtained.

(6) Before a student pilot is authorised to leave the circuit area on a solo flight to the general flying area or on a solo navigation flight, a flight instructor shall endorse the student's logbook in accordance with SA-CATS 61.

(7) A dual progress check flight must be conducted at the latest after a student completed his/her first 10 hours dual flight instruction and after each completion of 10 hours flight time thereafter.

(a) the dual progress check flight must be conducted by the Chief Flying Instructor (CFI) or by a Grade II or Grade I instructor appointed by the CFI.

(b) each dual progress check flight must be endorsed by the checking instructor in the logbook of the student in accordance with SA-CATS 61.

3. Solo flights in the General Training Area and solo navigation flights

(1) The student shall adhere to the authorised exercises while conducting solo flights in the general training area.

(2) The student shall adhere to the authorised route while conducting his/her solo navigation flight.

(3) The solo navigation flight must –

(a) include full-stop landings at two aerodromes away from base;

(b) have a total distance of not less than 150 nautical miles with a radius not exceeding 100 nautical miles from the base, along any sector of the flight.

(4) Except for the purpose of conducting the exercise 17b (precautionary landing), no flight below 500 feet above ground level (AGL) shall be conducted unless an instructor is on board the aircraft.

61.03.1 REQUIREMENTS FOR PRIVATE PILOT LICENCE (AEROPLANE)

1. Training

(1) Aim of training course

(a) The aim of the course is to train a candidate to the level of proficiency required for the issue of a private pilot licence (Aeroplane), and provide the training necessary to act, but not for remuneration, as pilot-in-command or as co-pilot of any aeroplane for which he or she holds a valid class or type rating, engaged in non-revenue flights under visual flight rules.

(2) Contents and requirements of training course

(a) The applicant must complete a training course with an aviation training organisation approved by the Director in terms of Part 141. However, the theoretical knowledge course and the practical training course may be completed at different aviation training organisations.

(b) The course comprises –

(i) theoretical knowledge course; and

(ii) practical training course.

(3) Theoretical knowledge course

(4) Theoretical knowledge course syllabus

(a) The detailed theoretical knowledge syllabus is contained in Appendix 1.0 to Document SA-CATS 61.

(b) The practical training syllabus is contained in Appendix 1.1 to Document SA-CATS 61.

(5) Radio Telephony

(a) To be eligible for the RTC a candidate must prove his knowledge of the ITU (International Telecommunications Union), and CAA requirements in both written and oral tests. The requirements are –

(i) ITU: Knowledge of radiotelephony operation and procedures, ability to transmit and receive messages by radio and knowledge of the radio communication regulations (e.g. use of different appropriate frequencies, interference, etc.) and especially those relating to the safety of human life.

(ii) ICASA: Proficiency in operating a radio installation including changing frequency, transmitting and receiving messages using the prescribed procedures, identification of Morse code beacons and clearing minor external faults.

(iii) The CAA: requirements are based on documents issued by the International Civil Aviation Organisation (ICAO) dealing with procedures, abbreviations and ing, the various aeronautical information circulars (AIC) in force, the S.A. Aeronautical Information Publication (AIP), Civil Aviation Regulations (CAR)

and Civil Aviation Technical Standards (CATS) and also other relevant regulations or rules in force.

(iv) Applicants for a Restricted Radio Certificate shall pass a theoretical Restricted Radio Examination at an approved CAA examination centre. The training syllabus for a Restricted Radio Certificate is contained in Appendix 1.5 to Document SA-CATS 61.

(b) Applicants for a Restricted Radio Certificate shall in addition to the theoretical knowledge examination pass a practical communication test including the identification of radio beacons using morse code conducted by a Designated Radio Examiner. The practical test shall include the full completion of an ATC Flight Plan and examine at minimum the following aspects:

(i) Use of Radio on the Ground.

(aa) Obtaining and complying with taxi instructions.

(bb) Knowing what to expect from the Air Traffic Control (ATC).

(cc) The importance of reading back all “hold short of instructions”.

(dd) The avoidance of runway incursions.

(ee) Meaning of “give way to other aircraft”.

(ff) Obtaining and complying with take-off instructions.

(gg) Importance of understanding “line up behind”.

(hh) Importance of reading back the take-off clearance.

(ii) Importance of reading back any other required instructions.

(jj) Radio procedures at unmanned/uncontrolled aerodromes.

(ii) Departure procedures.

(aa) Knowing what to expect in respect of departure procedures.

(bb) Required calls to be made on leaving the aerodrome circuit area.

(iii) En route procedures.

(aa) Knowing what call should be made to which station and when according to the airspace requirements.

(bb) Knowing the required in-flight broadcast procedure applicable to uncontrolled airspace.

(cc) Making a position report.

(dd) Obtaining relevant weather information, use of ATIS.

(ee) Making appropriate weather reports (PIREPS).

(ff) Knowing the difference between positively controlled airspace as opposed to a Flight Information Service (FIS).

(gg) Transponder use.

(hh) Procedure to obtain bearings, headings and position from air traffic control/FIS Altimeter setting procedures.

(ii) Relaying messages for other stations.

(iv) Arrival and traffic pattern procedures.

(aa) Knowing what to expect.

(bb) Arrival clearance/instructions.

(cc) Calls and ATC instructions whilst joining the traffic pattern.

(dd) Calls to be made in the circuit.

(ee) Calls to be made on vacating the runway.

(c). Applicants for a Restricted Radio Certificate shall attach a certificate of competency (Form CA 61.03.2a) signed by a Designated Radio Examiner when applying for a private pilot licence.

2. Theoretical Knowledge Examination

The knowledge acquired must be sufficient for the candidate to pass examinations in the following theoretical knowledge subjects –

(a) Aviation Meteorology (duration: 60 min);
(b) Flight Performance and Planning (duration: 90 min);
(c) General Navigation (duration: 90 min);
(d) Aircraft General (duration: 45 min);
(e) Principles of Flight (duration: 45 min);
(f) Human Performance and Limitations (duration: 45 min);
(g) Air Law (duration: 60min).

3. Practical Flight Test Standard

The Practical Flight Test Standard is found in Appendix 1.2 to Document SA-CATS 61

61.03.2 APPLICATION FOR AND ISSUE OF PRIVATE PILOT LICENCE (AEROPLANE)

1. Format of logbook

(1) The logbook summary shall be completed in the format indicated in Table 1 and submitted together with the application form.

Table 1

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
A/C Class	Instrument			Instructor		FSTD	Single Engine Day			
or Type	Actual Time	FSTD Time	SE	ME	FSTD		Dual	PIC	PICUS	Co-Pilot
PA 160							4.3	4.3		
C 172	3.0						15.2	12.9		
FNPT 1		3.8					18.0	4.3		
	3.0	3.8					37.5	21.5		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)

(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)
Single Engine Night				Multi Engine Day				Multi Engine Day			
Dual	PIC	PICUS	Co-Pilot	Dulal	PIC	PICUS	Co-pilot	Dual	PIC	PICUS	Co-pilot

(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)
Total Flight Time			59.0		Hours						

61.03.4 SKILLS TEST FOR PRIVATE PILOT LICENCE (AEROPLANE)

1. Procedures and manoeuvres

(1) The Skills Test shall be conducted in accordance with Appendix 1.2 to SA-CATS 61. Guidance to Designated Flight Examiner is provided on page 2 of the skills test form, Form CA 61-03.4, in respect of retesting. The skills test Form CA 61-03.4, completed by the Designated Flight Examiner, shall accompany the application form.

(2) The navigation element of the skills test administered for the issuing of a private pilot licence may be conducted as a separate flight within a maximum period of 14 days.

(3) The cross-country navigation flight of the skills test shall not be less than 200 nautical miles total distance and must include take-offs and landings at two aerodromes away from base. At least one of the aerodromes from which the aircraft takes off for this flight shall be an aerodrome at which an Air Traffic Services Unit (ATSU) is in operation.

61.04.1 REQUIREMENTS FOR PRIVATE PILOT LICENCE (HELICOPTER)

1. Training

(1). Aim of training course

(a) The aim of the course is to train a candidate to the level of proficiency required for the issue of a private pilot licence (Helicopter), and provide the training necessary to act, but not for remuneration, as pilot-in-command or as co-pilot of any helicopter for which he or she holds a valid class or type rating, engaged in non-revenue flights under visual flight rules.

(2) Contents and requirements of training course

(a) The applicant must complete a training course with an aviation training organisation approved by the Director in terms of Part 141. However, the theoretical knowledge course and the practical training course may be completed at different aviation training organisations.

(b) The course comprises –

(i) theoretical knowledge course; and

(ii) practical training course.

(3) Theoretical knowledge course

(4) Theoretical knowledge course syllabus

The detailed theoretical knowledge syllabus is contained in Appendix 1.0 to Document SA-CATS 61.

(5) Radio Telephony

(a) To be eligible for the RTC a candidate must prove his knowledge of the ITU (International Telecommunications Union), and CAA requirements in both written and oral tests. The requirements are:

(i) ITU: Knowledge of radiotelephony operation and procedures, ability to transmit and receive messages by radio and knowledge of the radio communication regulations (e.g. use of different appropriate frequencies, interference, etc. and especially those relating to the safety of human life.

(ii) ICASA: Proficiency in operating a radio installation including changing frequency, transmitting and receiving messages using the prescribed procedures, identification of Morse code beacons and clearing minor external faults.

(iii) The CAA: requirements are based on documents issued by the International Civil Aviation Organisation (ICAO) dealing with procedures, abbreviations and flight planning, the various aeronautical information circulars (AIC) in force, the S.A. Aeronautical Information Publication (AIP), Civil Aviation Regulations (CAR) and Civil Aviation Technical Standards (CATS) and also other relevant regulations or rules in force.

(iv) Applicants for a Restricted Radio Certificate shall pass a theoretical Restricted Radio Examination at an approved CAA examination centre. The training syllabus for a Restricted Radio Certificate is contained in Appendix 1.5 to Document SA-CATS 61.

(b) Applicants for a Restricted Radio Certificate shall in addition to the theoretical knowledge examination pass a practical communication test including the identification of radio beacons using morse code conducted by a Designated Radio Examiner. The practical test shall include the full completion of an ATC Flight Plan and examine at minimum the following aspects:

(i) Use of Radio on the Ground.

(aa) Obtaining and complying with taxi instructions.

(bb) Knowing what to expect from the Air Traffic Control (ATC).

(cc) The importance of reading back all “hold short of instructions”.

(dd) The avoidance of runway incursions.

(ee) Meaning of “give way to other aircraft”.

(ff) Obtaining and complying with take-off instructions.

(gg) Importance of understanding “line up behind”.

(hh) Importance of reading back the take-off clearance.

(ii) Importance of reading back any other required instructions.

(jj) Radio procedures at unmanned/uncontrolled aerodromes.

(ii) Departure procedures.

(aa) Knowing what to expect in respect of departure procedures.

(bb) Required calls to be made on leaving the aerodrome circuit area.

(iii) En route procedures.

(aa) Knowing what call should be made to which station and when according to the airspace requirements.

(bb) Knowing the required in-flight broadcast procedure applicable to uncontrolled airspace.

(cc) Making a position report.
(dd) Obtaining relevant weather information, use of ATIS.
(ee) Making appropriate weather reports (PIREPS).
(ff) Knowing the difference between positively controlled airspace as opposed to a Flight Information Service (FIS).
(gg) Transponder use.
(hh) Procedure to obtain bearings, headings and position from air traffic control/FIS Altimeter setting procedures.
(ii) Relaying messages for other stations.

(iv) Arrival and traffic pattern procedures.
(aa) Knowing what to expect.
(bb) Arrival clearance/instructions.
(cc) Calls and ATC instructions whilst joining the traffic pattern.
(dd) Calls to be made in the circuit.
(ee) Calls to be made on vacating the runway.

(c) Applicants for a Restricted Radio Certificate shall attach a certificate of competency (Form CA 61.03.2a) signed by a Designated Radio Examiner when applying for a private pilot licence.

2. Theoretical Knowledge Examination

The knowledge acquired must be sufficient for the candidate to pass examinations in the following theoretical knowledge subjects –

(a) Aviation Meteorology (duration: 60 min);
(b) Flight Performance and Planning (duration: 90 min);
(c) General Navigation (duration: 90 min);
(d) Aircraft General (duration: 45 min);
(e) Principles of Flight (duration: 45 min);
(f) Human Performance and Limitations (duration: 45 min);
(g) Air Law (duration: 60min).

3. Practical Flight Test Standard

The Practical Flight Test Standard is found in Appendix 1.2 to Document SA-CATS 61.

61.04.2 APPLICATION FOR AND ISSUE OF PRIVATE PILOT LICENCE (HELICOPTER)

1. Format of logbook

(1) The logbook summary shall be completed in the format indicated in Table 1 and submitted together with the application form.

Table 1

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
A/C Class	Instrument			Instructor		FSTD	Single Engine Day			
or Type	Actual Time	FSTD Time	SE	ME	FSTD		Dual	PIC	PICUS	Co-Pilot
R22							4.3	4.3		
R44	3.0						15.2	12.9		
FNPT 1		3.8					18.0	4.3		

	3.0	3.8					37.5	21.5		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)

(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)
Single Engine Night				Multi Engine Day				Multi Engine Day			
Dual	PIC	PICUS	Co-Pilot	Dulal	PIC	PICUS	Co-pilot	Dual	PIC	PICUS	Co-pilot
(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)
Total Flight Time			59.0		Hours						

61.04.4 SKILLS TEST FOR PILOT LICENCE (HELICOPTER)

1. Procedures and manoeuvres

The Skills Test shall be conducted in accordance with Appendix 1.4 to SA-CATS 61. Guidance to Designated Flight Examiner is provided on page 2 of the skills test form, in respect of retesting. The skills test form completed by the Designated Flight Examiner, shall accompany the application form.

61.05.1 REQUIREMENTS FOR COMMERCIAL PILOT LICENCE (AEROPLANE)

1. Training

(1) Aim of training course

The aim of the course is to train a candidate to the level of proficiency required for the issue of a Commercial Pilot licence (Aeroplane), and provide the training necessary to act, but not for remuneration, as pilot-in-command or as co-pilot of any aeroplane for which he or she holds a valid class or type rating, engaged in non-revenue flights under visual flight rules.

(2) Contents and requirements of training course

(a) The applicant must complete a training course with an aviation training organisation approved by the Director in terms of Part 141. However, the theoretical knowledge course and the practical training course may be completed at different aviation training organisations.

(b) The course comprises –

(i) theoretical knowledge course; and

(ii) practical training course. The syllabus for the practical training is contained in Appendix 2.1 to Document SA-CATS 61.

2. Theoretical Knowledge Examination

The knowledge acquired must be sufficient for the candidate to pass examinations in the following theoretical knowledge subjects:

(1) In the case of CPL VFR only:

(a) Aviation Meteorology;
(b) Flight Performance and Planning;
(c) Radio Aids and Communication;
(d) General Navigation;
(e) Instruments and Electronics;
(f) Aircraft Technical and General;
(g) Human Performance and Limitations;
(h) Air Law.

(2) In the case of CPL with IFR:

(a) Aviation Meteorology;
(b) Flight Performance and Planning;
(c) Radio Aids and Communication;
(d) General Navigation;
(e) Instruments and Electronics;
(f) Aircraft Technical and General;
(g) Human Performance and Limitations;
(h) Air Law and Operational Procedures.

3. Theoretical knowledge course syllabus

The detailed theoretical knowledge syllabus is contained in Appendix 2.0A to Document SA-CATS 61.

4. Radio Telephony

(1) To be eligible for the RTC a candidate must prove his knowledge of the ITU (International Telecommunications Union), and CAA requirements in both written and oral tests. The requirements are:

(a) ITU: Knowledge of radiotelephony operation and procedures, ability to transmit and receive messages by radio and knowledge of the radio communication regulations (e.g. use of different appropriate frequencies, interference, etc.) and especially those relating to the safety of human life.

(b) ICASA: Proficiency in operating a radio installation including changing frequency, transmitting and receiving messages using the prescribed procedures, identification of Morse code beacons and clearing minor external faults.

(c) The CAA: requirements are based on documents issued by the International Civil Aviation Organisation (ICAO) dealing with procedures, abbreviations and flight

planning, the various aeronautical information circulars (AIC) in force, the S.A. Aeronautical Information Publication (AIP), Civil Aviation Regulations (CAR) and Civil Aviation Technical Standards (CATS) and also other relevant regulations or rules in force.

(d) Applicants for a General Radio Certificate shall pass a theoretical General Radio Examination at an approved CAA examination centre. The training syllabus for a General Radio Certificate is contained in Appendix 1.5a to Document SA-CATS 61.

(2) Applicants for a General Radio Certificate shall in addition to the theoretical knowledge examination pass a practical communication test including the identification of radio beacons using morse code conducted by a Designated Radio Examiner (General). The practical test shall include the full completion of an ATC IFR Flight Plan and examine, at minimum, the following aspects for an IFR flight departing on a standard or non-standard departure from a controlled aerodrome, into a CTR/TMA, flying in both controlled and uncontrolled airspace en route to a controlled airfield as destination, complying with a standard arrival (STAR) and the completion of an instrument approach procedure (precision or non-precision):

(a) Use of Radio on the Ground –
- (i) Obtaining start clearance
- (ii) Obtaining taxi clearance
- (iii) Acceptance and read back of ATC departure clearance (Standard/non-standard)

(b) Departure procedure –
- (i) Take-off clearance
- (ii) Use of SID chart/compliance with non-standard departure procedure
- (iii) Selection of departure frequency and contact with relevant ATSU
- (iv) Use of area chart if applicable

(c) En route procedures –
- (i) Use of radio navigation chart
- (ii) Selection of frequencies appropriate to the route
- (iii) Passing and revising estimates
- (iv) Complying with onward clearance time (OCT)

(d) Arrival procedures –
- (i) Use of area chart if applicable
- (ii) Acceptance and review of STAR and instrument approach charts
- (iii) Radar vectors to ILS localiser or Holding beacon for non-precision approach including expected approach time (EAT)
- (iv) Establishing weather minima at landing aerodrome and compliance with Approach Ban
- (v) Completion of instrument approach procedure using IAC and correct radio procedures

(e) Radio communication failure –

Emphasis shall be placed on correct use of aviation words and phrases as well as the avoidance of slang terms.

61.05.2 APPLICATION FOR AND ISSUE OF COMMERCIAL PILOT LICENCE (AEROPLANE)

1. Format of logbooks

(1) The logbook summary shall be completed in the format indicated in Table 1 and submitted together with the application form.

Table 1

Logbook Summary

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
A/C Class or Type	Instrument			Instructor		FSTD	Single Engine Day			
	Actual Time	FSTD Time	SE	ME	FSTD		Dual	PIC	PICUS	Co-Pilot
PA 160							40.0	27.2		
C 172	15.0.0						53.3	67.2		
FNPT 1		21.0								
BE 55	5.0									
	20.0	21.0					93.3	94.4		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)

(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)
Single Engine Night				Multi Engine Day				Multi Engine Day			
Dual	PIC	PICUS	Co-Pilot	Dulal	PIC	PICUS	Co-pilot	Dual	PIC	PICUS	Co-pilot
12.0	4.5										
	3.0										
				9.1	2.0			2.0	1.0		
12.0	7.5			9.1				2.0	1.0		
(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)
Total Flight Time			219.3		Hours						

61.05.4 SKILLS TEST FOR COMMERCIAL PILOT LICENCE (AEROPLANE)

1. Procedures and manoeuvres

The practical test standard shall be conducted in accordance with Appendix 2.1 to SA-CATS 61.

61.06.1 REQUIREMENTS FOR COMMERCIAL PILOT LICENCE (HELICOPTER)

1. Training

(1) Aim of training course

The aim of the course is to train a candidate to the level of proficiency required for the issue of a Commercial Pilot licence (Helicopter), and provide the training necessary to act as pilot-in-command or as co-pilot of any helicopter for which he or she holds a valid class or type rating engaged in flights under visual flight rules.

(2) Contents and requirements of training course

(a) The applicant must complete a training course with an aviation training organisation approved by the Director in terms of Part 141. However, the theoretical knowledge course and the practical training course may be completed at different aviation training organisations.

(b) The course comprises –

(i) theoretical knowledge course; and

(ii) practical training course. The syllabus for the practical training is contained in Appendix 2.1 to Document SA-CATS 61.

(3) Theoretical Knowledge Examination

The knowledge acquired must be sufficient for the candidate to pass examinations in the following theoretical knowledge subjects:

(a) In the case of CPL VFR only:

(a) Aviation Meteorology;

(b) Flight Performance and Planning;

(c) Radio Aids and Communication;

(d) General Navigation;

(e) Instruments and Electronics;

(f) Aircraft Technical and General;

(g) Human Performance and Limitations;

(h) Air Law.

(b) In the case of CPL with IFR:

(i) Aviation Meteorology;

(ii) Flight Performance and Planning;

(iii) Radio Aids and Communication;

(iv) General Navigation;

(v) Instruments and Electronics;

(vi) Aircraft Technical and General;

(vii) Human Performance and Limitations;

(viii) Air Law and Operational Procedures.

(4) Theoretical knowledge course syllabus

The detailed theoretical knowledge syllabus is contained in Appendix 2.0A to Document SA-CATS 61.

(5) Radio Telephony

(a) To be eligible for the RTC a candidate must prove his knowledge of the ITU (International Telecommunications Union), and CAA requirements in both written and oral tests. The requirements are:

(i) ITU: Knowledge of radiotelephony operation and procedures, ability to transmit and receive messages by radio and knowledge of the radio

communication regulations (e.g. use of different appropriate frequencies, interference, etc.) and especially those relating to the safety of human life.

(ii) ICASA: Proficiency in operating a radio installation including changing frequency, transmitting and receiving messages using the prescribed procedures, identification of Morse code beacons and clearing minor external faults.

(iii) The CAA: requirements are based on documents issued by the International Civil Aviation Organisation (ICAO) dealing with procedures, abbreviations and flight planning, the various aeronautical information circulars (AIC) in force, the S.A. Aeronautical Information Publication (AIP), Civil Aviation Regulations (CAR) and Civil Aviation Technical Standards (CATS) and also other relevant regulations or rules in force.

(iv) Applicants for a General Radio Certificate shall pass a theoretical General Radio Examination at an approved CAA examination centre. The training syllabus for a General Radio Certificate is contained in Appendix 1.5a to Document SA-CATS 61.

(b) Applicants for a General Radio Certificate shall in addition to the theoretical knowledge examination pass a practical communication test including the identification of radio beacons using morse code conducted by a Designated Radio Examiner (General). The practical test shall include the full completion of an ATC IFR Flight Plan and examine, at minimum, the following aspects for an IFR flight departing on a standard or non-standard departure from a controlled aerodrome, into a CTR/TMA, flying in both controlled and uncontrolled airspace en route to a controlled airfield as destination, complying with a standard arrival (STAR) and the completion of an instrument approach procedure (precision or non-precision):

(i) Use of Radio on the Ground

(aa) Obtaining start clearance

(bb) Obtaining taxi clearance

(cc) Acceptance and read back of ATC departure clearance (Standard/non-standard)

(ii) Departure procedure

(aa) Take-off clearance

(bb) Use of SID chart/compliance with non-standard departure procedure

(cc) Selection of departure frequency and contact with relevant ATSU

(dd) Use of area chart if applicable

(iii) En route procedures

(aa) Use of radio navigation chart

(bb) Selection of frequencies appropriate to the route

(cc) Passing and revising estimates

(dd) Complying with onward clearance time (OCT)

(iv) Arrival procedures

(aa) Use of area chart if applicable

(bb) Acceptance and review of STAR and instrument approach charts

(cc) Radar vectors to ILS localiser or Holding beacon for non-precision approach including expected approach time (EAT)

(dd) Establishing weather minima at landing aerodrome and compliance with Approach Ban

(ee) Completion of instrument approach procedure using IAC and correct radio procedures

(v) Radio communication failure

Emphasis shall be placed on correct use of aviation words and phrases as well as the avoidance of slang terms.

61.06.2 APPLICATION FOR AND ISSUE OF COMMERCIAL PILOT LICENCE (HELICOPTER)

1. Format of logbooks

(1) The logbook summary shall be completed in the format indicated in Table 1 and submitted together with the application form.

Table 1

Logbook Summary

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
A/C Class	Instrument			Instructor		FSTD	Single Engine Day			
or Type	Actual Time	FSTD Time	SE	ME	FSTD		Dual	PIC	PICUS	Co-Pilot
R22							40.0	36.3		
R44	20.0.0						53.3	67.2		
FNPT 1		21.0								
	20.0	21.0					93.3	103.5		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)

(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)
Single Engine Night				Multi Engine Day				Multi Engine Day			
Dual	PIC	PICUS	Co-Pilot	Dulal	PIC	PICUS	Co-pilot	Dual	PIC	PICUS	Co-pilot
12.0	4.5										
1.0	3.0										

13.0	7.5			9.1				2.0	1.0		
(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)
Total Flight Time			217.8		Hours						

61.06.4 SKILLS TEST FOR COMMERCIAL PILOT LICENCE (HELICOPTER)

1. Procedures and manoeuvres

The practical test standard shall be conducted in accordance with Appendix 2.2 to SA-CATS 61.

61.07.1 REQUIREMENTS FOR AIRLINE TRANSPORT PILOT LICENCE (AEROPLANE)

1. Training

(1) Aim of training course

The aim of the course is to train a candidate to the level of proficiency required for the issue of an Airline Transport Pilot licence (Aeroplane), and provide the training necessary to act, but not for remuneration, as pilot-in-command or as co-pilot of any aeroplane for which he or she holds a valid class or type rating, engaged in non-revenue flights under visual flight rules.

(2) Contents and requirements of training course

(a) The applicant must complete a training course with an aviation training organisation approved by the Director in terms of Part 141. However, the theoretical knowledge course and the practical training course may be completed at different aviation training organisations.

(b) The course comprises –

(i) Theoretical knowledge course; and

(ii) Practical training course is only applicable to the Integrated Training Course. The syllabus for the practical training is contained in Appendix 3.0 to Document SA-CATS 61.

(3) Theoretical Knowledge Examination

(a) The knowledge acquired must be sufficient for the candidate to pass examinations in the following theoretical knowledge subjects:

(i) Aviation Meteorology;
(ii) Flight Performance and Planning;
(iii) Radio Aids and Communication;
(iv) General Navigation;
(v) Instruments and Electronics;
(vii) Aircraft Technical and General;

(b) In the case of a student pilot following an integrated course or a private pilot, the following two exams will also have to be written:

(i) Human Performance and Limitations;
(ii) Air Law and Operational Procedures;
(iii) Navigation (at the CPL(IF) level).

(4) Theoretical knowledge course syllabus

The detailed theoretical knowledge syllabus is contained in Appendix 2.0B to Document SA-CATS 61.

61.07.2 APPLICATION FOR AND ISSUE OF AIRLINE TRANSPORT PILOT LICENCE (AEROPLANE)

1. Format of logbooks

(1) The logbook summary shall be completed in the format indicated in Table 1 and submitted together with the application form.

Table 1

Logbook Summary

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
A/C Class or Type	Instrument			Instructor		FSTD	Single Engine Day			
	Actual Time	FSTD Time	SE	ME	FSTD		Dual	PIC	PICUS	Co-Pilot
PA 160							40.0	27.2		
C 172	15.0						53.3	67.2		
FNPT 1		21.0								
BE 55	5.0									
C208	29.0						12.0	250.7	120.2	
BE1900	37.8	17.0								
C551	17.0	34.4								
B737	103.2	65.0								
	276.8	103.0					105.3	345.1	120.2	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)

(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)
Single Engine Night				Multi Engine Day				Multi Engine Day			
Dual	PIC	PICUS	Co-Pilot	Dulal	PIC	PICUS	Co-pilot	Dual	PIC	PICUS	Co-pilot
12.0	4.5										
	3.0										
				9.1	2.0			2.0	1.0		
2.0	50.7	20.2	138.4								
				7.0	237.0	120.0	572.0	3.1	9.4	2.7	15.8
				4.0	120.0	60.0	420.0	2.0	4.7	3.2	45.8
				12.0		230.0	952.0	10.0		15.0	158.0

14.0	58.2	20.2	138.4	32.1	359.0	410.0	1944.0	17.0	15.1	20.9	219.6
(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)
Total Flight Time			3809.1		Hours						

61.07.4 SKILLS TEST FOR AIRLINE TRANSPORT PILOT LICENCE (AEROPLANE)

1. Procedures and manoeuvres

The practical test standard shall be conducted in accordance with Appendix 9.0 to SA-CATS 61.

61.08.1 REQUIREMENTS FOR AIRLINE TRANSPORT PILOT LICENCE (HELICOPTER)

1. Training

(1) Aim of training course

The aim of the course is to train a candidate to the level of proficiency required for the issue of an Airline Transport Pilot licence (Helicopter), and provide the training necessary to act, but not for remuneration, as pilot-in-command or as co-pilot of any helicopter for which he or she holds a valid class or type rating, engaged in non-revenue flights under visual flight rules.

(2) Contents and requirements of training course

(a) The applicant must complete a training course with an aviation training organisation approved by the Director in terms of Part 141. However, the theoretical knowledge course and the practical training course may be completed at different aviation training organisations.

(b) The course comprises –

(i) theoretical knowledge course; and

(ii) practical training course is only applicable to the Integrated Training Course. The syllabus for the practical training is contained in Appendix 3.0 to Document SA-CATS 61.

(3) Theoretical Knowledge Examination

(a) The knowledge acquired must be sufficient for the candidate to pass examinations in the following theoretical knowledge subjects:

(i) Aviation Meteorology;
(ii) Flight Performance and Planning;
(iii) Radio Aids and Communication;
(iv) General Navigation;
(v) Instruments and Electronics;
(vi) Aircraft Technical and General.

(b) In the case of a student pilot following an integrated course or a private pilot, the following two exams will also have to be written:

(i) Human Performance and Limitations;
(ii) Air Law and Operational Procedures
(iii) Navigation (at the CPL (IF) level).

(4) Theoretical knowledge course syllabus

The detailed theoretical knowledge syllabus is contained in Appendix 2.0B to Document SA-CATS 61.

61.08.2 APPLICATION FOR AND ISSUE OF AIRLINE TRANSPORT PILOT LICENCE (HELICOPTER)

1. Format of logbooks

(1) The logbook summary shall be completed in the format indicated Table 1 and submitted together with the application form.

Table 1

Logbook Summary

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
A/C Class	Instrument			Instructor		FSTD	Single Engine Day			
or Type	Actual Time	FSTD Time	SE	ME	FSTD		Dual	PIC	PICUS	Co-Pilot
R22							40.0	27.2		
R44	15.0						53.3	67.2		
FNPT 1		21.0								
Bell 206LT	5.0									
Bell47T	29.0						12.0	250.7	120.2	
AS332	37.8	17.0								
SA330	17.0	34.4								
SK 61	103.2	65.0								
	276.8	103.0					105.3	345.1	120.2	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)

(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)
Single Engine Night				Multi Engine Day				Multi Engine Day			
Dual	PIC	PICUS	Co-Pilot	Dulal	PIC	PICUS	Co-pilot	Dual	PIC	PICUS	Co-pilot
12.0	4.5										
	3.0										
				9.1	2.0			2.0	1.0		
2.0	50.7	20.2	138.4								
				7.0	237.0	120.0	572.0	3.1	9.4	2.7	15.8
				4.0	120.0	60.0	420.0	2.0	4.7	3.2	45.8
				12.0		230.0	952.0	10.0		15.0	158.0

14.0	58.2	20.2	138.4	32.1	359.0	410.0	1944.0	17.0	15.1	20.9	219.6
(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)
Total Flight Time			3809.1		Hours						

61.08.4 SKILLS TEST FOR AIRLINE TRANSPORT PILOT LICENCE (HELICOPTER)

1. Procedures and manoeuvres

The practical test standard shall be conducted in accordance with Appendix 9.0 to SA-CATS 61.

61.09.1 REQUIREMENTS FOR ISSUE OF CLASS AND TYPE RATINGS

1. Licence endorsements

The list of class and type ratings for endorsement are set out in Table 1 in TS 61.09.8.

2. Class and Type Rating Training

(1) Theoretical knowledge instruction requirements for class/type ratings

(a) The theoretical knowledge instruction shall be conducted by competent persons having appropriate experience in aviation and knowledge of the aircraft concerned, eg. flight instructor, flight engineer, maintenance engineer.

(b) The theoretical knowledge instruction shall cover the syllabus in Appendix 8.0 to SA-CATS 61, as appropriate to the aeroplane class/type concerned. Depending on the equipment and systems installed, the instruction shall include but is not limited to the following content

(i) Aeroplane structure and equipment, normal operation of systems and malfunctions

(aa) Dimensions;
(bb) Engine including auxiliary power unit;
(cc) Fuel system;
(dd) Pressurisation and air-conditioning;
(ee) Ice protection, windshield wipers and rain repellent;
(ff) Hydraulic systems;
(gg) Landing gear;
(hh) Flight controls, lift devices;
(ii) Electrical power supply;
(jj) Flight instruments, communication, radar and navigation equipment;
(kk) Cockpit, cabin and cargo compartment;
(ll) Emergency equipment.

(ii) Limitations

(aa) General limitations;
(bb) Engine limitations;
(cc) System limitations;
(dd) Minimum equipment list.

(iii) Performance, flight planning and monitoring

(aa) Performance;
(bb) Flight planning;
(cc) Flight monitoring.

(iv) Load, balance and servicing

(aa) Load and balance;
(bb) Servicing on ground.

(v) Emergency procedures

(vi) Special requirements for extension of a type rating for instrument approaches down to a decision height of less than 200 ft (60 m)

(aa) Airborne equipment, procedures and limitations.

(vii) Special requirements for “glass cockpit” aeroplanes

(aa) Electronic flight instrument systems (eg. EFIS, EICAS).

(viii) Flight Management systems (FMS)

(2) High Performance Aeroplane (HPA) Training Course of additional theoretical knowledge training for a class or type rating for high performance single-pilot aeroplane and warbird type endorsements.

(a) A number of aeroplanes certificated for single-pilot operation have similar performances, systems and navigation capabilities to those more usually associated with multi-pilot types of aeroplanes, and regularly operate within the same airspace. The level of knowledge required to operate safely in this environment is not part of, or not included to the necessary depth of knowledge in the training syllabi for the PPL (A), CPL (A) or IR (A) but these licence holders may fly as pilot-in-command of such aeroplanes.

(b) The aim of the theoretical knowledge course is to provide the applicant with sufficient knowledge of those aspects of the operation of aeroplanes capable of operating at high speeds and altitudes, and the aircraft systems necessary for such operation.

(c) The holder of an ATPL (A) issued by a Contracting State or with a pass in the theoretical knowledge examinations at ATPL (A) level is credited with meeting the requirement of CAR 61.09.4(3).

(d) A pass in any theoretical knowledge subjects as part of the HPA course will not be credited against meeting future theoretical examination requirements for issue of a CPL (A), IR (A) or ATPL (A).

(3). HPA Course Providers

Theoretical knowledge instruction for the HPA may be provided by a Part 141 approved aviation training organisation accredited to conduct theoretical knowledge training for the ATPL (A). Course providers will be required to certify completion of the training and demonstration of knowledge by the applicant as a pre-requisite for training for an initial type or class rating for aeroplanes designated as high performance.

(4) HPA Course Syllabus

There is no mandatory minimum or maximum duration of the theoretical knowledge instruction required for the HPA syllabus. The course material may be conducted by distance learning. The subjects to be covered in the course and written examination are given in Document SA-CATS .

(5) Multi-crew Co-operation Course

(a) The aim of the course is to become proficient in multi-crew co-operation (MCC) in order to operate safely multi-pilot multi-engine aircraft under IFR and, for that purpose, to ensure that–

(i) The pilot-in-command fulfils his managing and decision-making functions irrespective whether he is pilot flying (PF) or pilot not flying (PNF);

(ii) The tasks of PF and PNF are clearly specified and distributed in such a manner that the PF can direct his full attention to the handling and control of the aircraft;

(iii) Co-operation is effected in an orderly manner appropriate to the normal, abnormal or emergency situations encountered; and

(iv) Mutual supervision, information and support are ensured at all times.

(6) Instructors

Instructors for MCC training shall be approved as instructors for MCC training. They should be current with the latest developments in human factors training and CRM techniques.

(7) Theoretical Knowledge

The theoretical knowledge syllabus is set out in Appendix 20.0 to Document SA-CATS 61. An approved MCC theoretical knowledge course shall comprise not less than 25 hours.

(8) Flying Training

The flying training syllabus is set out in Appendix 20.0 to Document SA-CATS 61.

(9) Certificate of Completion

On completion of the MCC training course or upon completion of an initial multi-pilot aircraft type endorsement, the applicant shall be issued with a certificate of satisfactory course completion as set out in the appropriate form.

(10) Cross-Crediting

A holder of a certificate of completion of MCC training on aeroplanes or helicopters shall be exempted from the requirement to complete the theoretical knowledge syllabus, as set out in Appendix 20.0 to Document SA-CATS 61, in the event that applicant seeks multi-crew authorisation on an alternative category of aircraft.

(11) HPA Examination

(a) Demonstration of acquisition of the HPA knowledge will be undertaken by passing an examination set by the training provider and acceptable to the Director. Successfully passing this examination will result in the issue of a certificate indicating that the course and examination have been completed. The syllabus content for theoretical training for High Performance Aircraft is set out in Appendix 8.2 to the SA-CATS 61.

(b) The certificate will represent a 'once only' qualification and will satisfy the requirement for the addition of all future high performance aeroplanes to the holder's licence. The certificate will be valid indefinitely and must be submitted with the application of the first HPA type or class rating.

(c) The written examination shall consist of not less than 60 multi-choice questions, and may be split into individual subject papers at the discretion of the Part 141 approved aviation training organisation. The pass mark for the examination will be 75%.

(12) Class and Type Rating Examinations

(a) Demonstration of acquisition of the class or type rating knowledge will be undertaken by passing an examination(s) set by the Part 141 training provider or an equivalent organisation acceptable to the Director. Successfully passing this examination will result in the issue of a certificate indicating that the course and examination have been completed.

(b) For the initial issue of type ratings for multi-pilot aeroplanes the written or computer based examination shall at least comprise one hundred questions distributed appropriately across the main subjects of the syllabus. The pass mark shall be 75% in each of the main subjects of the syllabus.

(c) For the initial issue of type and class ratings for single-pilot aeroplanes the number of questions in the written or computer based examination shall depend on the complexity of the aeroplane. The pass mark shall be 75%.

(d) After the examination, the candidate shall review the questions answered incorrectly, with the instructor who provided the theoretical knowledge training, to correct the knowledge deficiency.

61.09.2 TRAINING

1. Training and prior qualifications

The training and prior qualification requirements for the issuing of class and type ratings (including high performance aeroplanes) are set out in Table A.

Table A

61.09.2 TRAINING

1. Training and prior qualifications

	SE Class	ME Class	ME SP Type	Multi-pilot	Warbird	Helicopter SE	Helicopter ME

Applicable Tables *	1, 2	1	3, 4, 5	6	8	7	7
Instrument rating (ME)	N/A	N/A	N/A	Required	N/A	N/A	N/A
MCC training	N/A	N/A	N/A	Required	N/A	N/A	N/A
ATPL theory credit	N/A	N/A	N/A	Required	N/A	N/A	N/A
Theoretical knowledge	App 8.0	App 8.0	App 8.0	App 8.0	App 8.0	App 8.3	App 8.3
Flight training	App 10.0	App 10.2	App 10.2	App 10.3	App 10.4	App 10.5	App 10.6

* The tables are contained in SA-CATS 61.09.8

2. General

(1) The type rating course, including theoretical knowledge, shall be completed within the 3 months preceding the skills test.

(2) Where the type is new to the aircraft register or a manufacturer requires specific type training, then the training done through that manufacturer or TRTO shall be acceptable as part of the type training requirements as referred to above.

(3) Where the type is new to the aircraft register and the pilot is able to present a FAA/JAR skills test carried out by a pre-approved examiner and where that skills test includes the oral exam pertaining to that regulatory authority, then that skills test will be deemed as being compliant with the Civil Aviations Regulations as specified below for the initial rating only.

61.09.3 SKILLS TESTS

Skills tests, the details of which can be found in Appendix 9.0, 'Guidance for the Conducting of Skills and Proficiency Tests' together with Appendices 9.1, 9.2 and 9.3 to Document SA-CATS 61, shall comprise the relevant elements from the following areas of operation

1. Normal Procedures

(1) Mass and balance data;
(2) Take-off and landing distance requirements;
(3) Altitude capability/flight planning;
(4) Weather interpretation;
(5) Filing of flight plan;
(6) Pre-flight inspection;
(7) Pre-start checks;
(8) Starting procedures after start procedures;
(9) Taxiing checks;
(10) Pre- take-off procedures and checks;
(11) Crew/pilot briefing;
(12) Departure procedures;
(13) Climb procedures including best rate/maximum angle and cruise climb techniques and engine monitoring procedures;
(14) Cruise techniques;
(15) Use of navigation systems;
(16) Use of automation;

(17) Descent techniques;
(18) Approach preparation and briefings;
(19) Flying the approach and relevant procedures;
(20) Landing techniques;
(21) After-landing procedures;
(22) Shutdown procedures; and
(23) Paperwork requirements.

2. Non-normal Procedures

(1) Operation on wet or contaminated runways;
(2) Operation in strong crosswinds;
(3) Operation in icing conditions;
(4) Windshear recovery techniques;
(5) Aborted and alternate engine start procedures;
(6) Rejected take-off;
(7) Engine failure procedures;
(8) System failure procedures;
(9) Instrument failure procedures;
(10) Avionic failure procedures;
(11) Radio failure procedures; and
(12) Declaring an emergency.

3. Crew Procedures

(1) CRM;
(2) Threat and error identification and management;
(3) Multi-crew co-operation;
(4) Communication including ATC communications;
(5) General management of the flight;
(6) Situational awareness.

4. Theoretical knowledge instruction and checking requirements

An applicant for a class or type rating for single- or multi-engine aeroplanes shall have completed the required theoretical knowledge instruction and demonstrated the level of knowledge required for the safe operation of the applicable aeroplane type.

5. Flight instruction

(1) An applicant for a class/type rating for single-engine and multi-engine single-pilot aeroplanes shall have completed a course of theory and flight instruction related to the class/type rating skills test.
(2) An applicant for a type rating for multi-pilot aeroplanes shall have completed a course of theory and flight instruction related to the type rating skills test.

6. Conduct of training courses

(1) Training courses shall be conducted by an Aviation Training organisation approved by the Director in terms of Part 141.
(2) Such courses shall be approved by the Director.

7. Multi-crew co-operation training

(1) The course shall provide multi-crew co-operation training for –
  (a) students attending an integrated course for an airline transport pilot licence in accordance with CAR 61.01.16 (2); or
  (b) the holders of a private pilot licence with instrument rating or commercial pilot licence with instrument rating, who have not graduated from an airline transport pilot integrated course but who wish to obtain an initial type rating on multi-pilot aeroplanes.

(2) The multi-crew co-operation course shall comprise at least 25 hours of theoretical knowledge instruction and exercises and 20 hours of multi-crew co-operation training. Students attending an airline transport pilot integrated course may have the practical training reduced by 5 hours. Wherever possible, the multi-crew co-operation training should be combined with the initial type rating course on multi-pilot aeroplanes or helicopters.

(3) The multi-crew co-operation training shall be conducted by an aviation training organisation approved by the Director in terms of Part 141. When multi-crew co-operation training is combined with the initial type rating training for a multi-pilot aeroplane, the practical multi-crew co-operation training may be reduced to not less than 10 hours provided the same FSTD is used for both the multi-crew co-operation and type rating training.

8. Additional training requirements for type or class ratings on High Performance Single-pilot Aeroplanes

The additional training requirements for an applicant for a class or type rating on a high performance single-pilot aeroplane and who is not the holder of an Airline Transport Pilot's Licence or holds credit for the ATP theoretical knowledge examinations shall be as set out in Appendix 8.2 to SA-CATS 61.

9. Warbird Qualification and Experience

The following qualifications and aeronautical experiences apply to warbird types of aeroplane fitted with dual controls –

(1) The appropriate aeroplane class rating;

(2) Any applicable design feature endorsements;

(3) For gas turbine engine powered warbird types of aeroplane not capable of exceeding Mach 1 in level flight, aeronautical experience of at least –

(a) 300 hours of flight time as pilot-in-command in aeroplanes; or

(b) 30 hours of flight time as pilot-in-command in gas turbine engine powered aeroplanes.

(4) For gas turbine engine powered warbird types of aeroplane capable of exceeding Mach 1 in level flight, aeronautical experience of at least 30 hours of flight time as pilot-in-command in turbojet or turbofan powered aeroplanes with a MMO of at least Mach 0.8;

(5) In addition, for a warbird type not fitted with dual controls:

(a) for piston engine powered type, at least 30 hours of flight time as pilot-in-command in aircraft having engine power in excess of 450 hp;

(b) for a gas turbine powered type not capable of exceeding Mach 1, at least 30 hours of flight time as pilot-in-command in gas turbine powered aeroplanes;

(c) for a gas turbine powered warbird type capable of exceeding Mach 1 in level flight, a minimum of 50 hours of flight time in gas turbine powered aeroplanes capable of exceeding Mach 1 in level flight;

(d) for a warbird with a delta wing, at least 15 hours of flight time as pilot-in-command in aeroplanes fitted with a delta wing;

(6) For multi-engine warbird aeroplane types, the aeronautical experience (except that mentioned in subparagraph (e) (iv) above) must be in multi-engine aeroplanes.

(7) Multi-pilot Skills Test. An applicant for a type rating for a multi-pilot aeroplane shall have demonstrated to a Designated Flight Examiner the skills required for the safe operation of the applicable type of aeroplane in a multi-crew environment as a pilot-in-command or a co-pilot as applicable, as set out in Appendices 9.0 and 9.1 to Document SA-CATS 61.

(8) Multi-engine Class Rating. An applicant for the issuing of a multi-engine class rating shall have demonstrated to a Designated Flight Examiner the competence to perform as pilot-in-command of the aircraft concerned the procedures and manoeuvres as described in Appendix 9.2 to Document SA-CATS 61.

(9) Single-engine Class or Type Rating. An applicant for the issuing of a single-engine class, type rating or touring motor glider class rating shall have demonstrated to a Designated

Flight Examiner or an appropriately rated flight instructor the competence to perform as pilot-in-command of the aircraft concerned the procedures and manoeuvres as described in Appendix 9.2 to Document SA-CATS 61.

(10) Warbird Type Rating. An applicant for the issuing of a warbird type rating shall have demonstrated to a Designated Flight Examiner or an appropriately rated flight instructor the competence to perform as pilot-in-command of the aircraft concerned the procedures and manoeuvres as described in Appendix 9.3 to Document SA-CATS 61.

(11) Multi-crew Co-operation. On completion of the MCC training the applicant shall either demonstrate the ability to perform the duties of a pilot on multi-pilot aeroplanes by passing the type rating skills test on multi-pilot aeroplanes as set out in Appendices 9.0 and 9.1 to SA-CATS 61, or shall be given a certificate of completion of MCC and have the successful completion of the course endorsed in the logbook.

(12) The skills test shall have been completed within 6 months of the date of completion of the training course.

61.09.7 TYPE AND CLASS RATING – PRIVILEGES AND VARIANTS

1. Difference training

The differences training required is indicated in Tables 1 – 10 in Technical Standard 61.09.8. In the case of a required change to another type or variant of the aeroplane within one class rating, the following will apply:

(a) Although an applicant will have an endorsement in his licence for a class rating in his/her licence, differences training will have to be conducted as defined below, endorsed into the pilot logbook and the form forwarded to the Director within 30 days of completion of the training:

(b) Theoretical and Flying Training

(i) A pilot undergoing differences training shall have completed a course of theory instruction relevant to the aircraft.

(ii) A pilot undergoing differences training shall have completed flight instruction related to the aircraft and shall include but is not limited to:

(aa) A minimum of five take-offs and landings;

(bb) Upper air work appropriate to the handling characteristics and different, or more complex systems pertaining to the aircraft.

(iii) Differences training shall be conducted by an aviation training organisation approved by the Director in terms of Part 141.

61.09.8 TYPE AND CLASS RATINGS

1. Issuing of class and type ratings

Class and Type ratings shall be issued as an endorsement in the pilot's logbook and licence on submission of the appropriate form, which shall be forwarded to the Director within 30 days of completion of the training. The endorsement in the logbook shall contain the following particulars:

(a) an indication of the type or class of aircraft in respect of which the endorsement is made;

(b) the type (and variant if applicable) and registration marks of the aircraft in which the skills test referred to in CAR 61.09.5, was performed;

(c) the name, licence number, designation and signature of the person making the endorsement;

(d) the date.

2. Establishment of Type Ratings

Criteria: For the establishment of type ratings for aeroplanes other than those included in Tables 4 – 8, all of the following shall be considered –

(a) airworthiness type certificate;

(i) handling characteristics;

(ii) certificated minimum flight crew complements; and
(iii) level of technology.

(b) High performance single-pilot aeroplanes.

Criteria: For the establishment of a class or type rating of a single-pilot aeroplane designated as high performance, all the following shall be considered –

(i) type of power plant;
(ii) provision and capabilities of airframe systems;
(iii) cabin pressurisation;
(iv) capabilities of navigation systems;
(v) performance both airfield and en route;
(vi) handling characteristics.

3. List of Classes and Types of aeroplanes

(1) Class of Aeroplane

Explanation of how to use Tables 1, 2 and 3:

(a) the symbol (D) in column 3 indicates that differences training is required when moving between variants or other types of aeroplane which are separated by the use of a line in column 2;

(b) Although the licence endorsement (column 4) contains all aeroplanes listed in column 2, the required familiarisation or differences training has still to be completed and endorsed in the pilot's logbook by the instructor or SACAA authorised person;

(c) The symbol HPA (High Performance Aeroplane) in column 3 indicates that additional knowledge instruction is required for this type of aeroplane if the applicant for the type rating is not the holder of an ATPL (A) or has no theoretical knowledge credit at ATPL (A) level.

Note: Aeroplanes not listed may be entered into a SA CAR 61 licence, but the rating privileges are restricted to aeroplanes on the SA register. At the time of publication of these Technical Standards every attempt was made to ensure that the information in respect of types was correct. Corrections and new entries will be made available on the SA CAA website www.caa.co.za. The tables will be updated from time to time.

Table 1

Single/multi-engine piston aeroplanes land/sea – Single-pilot (SP) (A)

1 Manufacturer	2 Aeroplanes	3	4 Licence Endorsement
All Manufacturers	Single-engine piston (land)	(D)	SEP (land)
	Single-engine piston (land) with variable pitch propellers (VP)		
	Single-engine piston (land) with Retractable undercarriage (RU)		
	Single-engine piston (land) with Turbo/super charged engines (T)		
	Single-engine piston (land) with cabin pressurisation (P)		
	Single-engine piston (land) with Tail wheel (TW)		
	Single-engine piston (sea)		

1 Manufacturer	2 Aeroplanes	3	4 Licence Endorsement
	Single-engine piston (sea) variable pitch propellers (VP)	(D)	SEP (sea)
	Single-engine piston (sea) with Turbo/super charged engines (T)		
	Single-engine piston (sea) with cabin pressurisation (P)		
	Multi-engine piston (land)	(D)	MEP (land)
	Multi-engine piston (sea)	(D)	MEP (sea)

Table 2
Single-engine turboprop (land) – Single-pilot

1 Manufacturer	2 Aeroplanes	3	Licence Endorsement
Aerospatiale (Socata)	TBM 700	(HPA)	Aerospatial SET
Snow/Rockwell/ Ayres	S2R turbo thrush		Snow/Ayres SET
Cessna	206 A/T Soloy 207 A/T Soloy	(D)	Cessna SET
	208		
De Havilland (Air Tech Canada) (Bombadier)	DHC-3 Turbo-Otter		DHC3
	DHC-2 Turbo-Beaver		DNC2
Gulfstream	AM.G-164D		Gulfstream SET
Pilatus	PC06 Series PC6 82H2	(D)	Pilatus SET
	PC-7		
Rhein Flugzeugbau	FT 600		Rhein Flugzeugbau SET

Table 3
Single-engine piston touring motor gliders (land) – Single-pilot

1 Manufacturer	2 Aeroplanes	3	Licence Endorsement
All Manufacturers	All touring Motor Gilders having integrally mounted non-retractable engine and a non-retractable propeller		TMG

(2) List of aeroplane types – Tables 4 to 8

These tables include aeroplanes type certificated under FAR/JAR 25, FAR/JAR 23, FAR/JAR 23 Commuter Category, FAR/JAR 25, BCAR or AIR 2051, and aeroplanes type certificated by another competent authority acceptable to the Director, but does not include –

(a) aeroplanes not type certificated in accordance with FAR/JAR 23, FAR/JAR 23 Commuter Category, FAR/JAR 25, BCAR or AIR 2051;

(b) aeroplanes type certificated in South Africa under special registration such as military, ex-military, experimental (NTCA) or vintage aeroplanes.

(c) Aeroplanes not listed in the tables may be entered into a South African licence subject to any specific requirements that may be laid down by the Director from time-to-time.

(d) Explanation of tables:

(i) the symbol (D) in column 3 indicates that differences training is required when moving between variants or to other types of aeroplane which are separated by the use of a line in column 2.

(ii) although the licence endorsement (column 4) contains all aeroplanes listed in column 2, the required familiarisation or differences training has still to be completed.

(iii) the specific variant on which the skills test for the type of rating has been completed will be recorded according to TS 61.09.10.

(iv) the symbol HPA (High Performance Aeroplane) in column 3 indicates that additional knowledge instruction is required for this type of aeroplane if the applicant for the type rating is not the holder of an ATPL (A) or has no theoretical knowledge credit at ATPL (A) level.

A: Single-pilot aeroplanes

Table 4

Multi-engine turboprop aeroplane (land) – single-pilot (SP) (A)

1 Manufacturer	2 Aeroplanes	3	Licence Endorsement
Asla GAF	Nomad - 22B 24A		AslaMET
Beechcraft	90 series	(HPA) (D)	BE90.99.100/200
	99 series		
	100 series		
	200 series		
	300 series	(HPA) (D)	BE300/1900
	1900 series		
Cessna-Rheims Aviation	F406 425	(HPA)	C406/425
	441		C441
De Havilland Canada (Bombardier)	DHC6 Series		DHC6
Dornier	DO-128-6		D128
	DO 228 series		D228
Embraeer	Bandeirante EMB 110		EMB110
Grumman	Tracker S2FT		S2FT
Mitsubishi	MU2B series	(HPA)	MU2B
Piaggio	P165		Piaggio 165
	P180	(HPA)	Piaggio 180
Pilatus Britten	BN2T Turbine Islander	(D)	BN2T
	BN2T-4R MSSA BN2T-4S Defender		
Piper	PA31 series Cheyenne	(HPA)	PA31/42

1 Manufacturer	2 Aeroplanes	3	Licence Endorsement
	PA42 series Cheyenne	(D)	
Rockwell	AC680 T AC690 series AC900 series	(HPA)	PA31/42
Short (Bombardier)	SC7Skyvan		SC7Skyvan
Swearingen/ Fairchild	226 T 226 T (B)	HPA (D)	SA226/227
	226AC 226TC		
	227TF		
	227AC 227AT 227BC		

Table 5
Single-engine – single-pilot

1 Manufacturer	2 Aeroplanes	3	Licence Endorsement
Pilatus	PC-7MkII PC-9P C-9(M)	(HPA)	FC9/PC7MkII
	PC12 series	(HPA)	
Piper	PA-46 Malibu PA-46 Malibu Turbine	HPA (D)	PA-15
Weider Extra	Extra 400	(HPA)	Extra 400

Table 6
Multi-engine turbo-prop (sea) – single-pilot

1 Manufacturer	2 Aeroplanes	3	Licence Endorsement
Canadair (Bombardier)	CL215T		CL215T

Table 7
Multi-engine turbo-jet (land) – single-pilot (SP)

1 Manufacturer	2 Aeroplanes	3	Licence Endorsement
Aerospatiale	MS 760 Paris	(HPA)	S760
Cessna	C501/500SP	(HPA) (D)	C501/551
	C651/500SP*		
	C525	(HPA)	C525

B: Multi-pilot aeroplanes
Table 8

1 Manufacturer	2 Aeroplanes	3	Licence Endorsement
	SN601 Corvette		SN601
Aerospatiale/Sud Aviation	SE210III IIIR VIN	(D)	SE210/10B3/11/12
	SE 10B3		
	SE 11		
	SE 12		
Aerospatiale/BAC	Concorde		Concorde
Aerospatiale/Nord Aviation	Nordiatlas 2501		ND25
	C160 P Transall		ND16
	260A Nord 262A-B-C Nord		ND 26
Aero-Spaceline	377 SGTF Super Guppy		Super Guppy
Airbus	A300-B1 B2 series 4 series C4-200 series F4-200 series		A300
	A300-FFCC		A300FFCC
	A310-100 series 300 series A300-B4 600 series C4 600 series F4 600 series		A310/300-600
	A318 A319-100 series A320-100 series 200 series A321-100 series 200 series		A320
	A330-300 series 200 series		A330
	A340-200 series 300 series 500 series 600 series		A340
	A300 – 600 series ST/Beluga		A300-600ST
ATR	ATR42 200/30/400	(D)	ATR42.72
	ATR42 500 72 Series		
Mitsubshi/ Beech/Raytheon	Beechjet 400 series MU300		Beech400/MU300
	B707-100 series 200 series	(D)	B707/720
	B720		
	B717 series		B717

Boeing	B727-100 series 200 series		B727
	B737-100 series 200 series		B737 100-200
	B737-300 series 400 series 500 series	(D)	B737 300-900
	600 series 700 series 800 series 900 series		
	B747-100 series 200 series 300 series	(D)	B747 100-300
	B747SP		
	B747-400 series		B747-400
	B757-200 series 300 series	(D)	B757/767
	767-200 series 300 series		
	B767-400ER*		
	B777-200 series 300 series		B777

* The differences training course is valid from the B757/767 ‘classic’ to the B767-400ER for crew members previously qualified on the B757/767 ‘classic variants’. The 767-400ER to B757/767 ‘classic’ differences training shall be evaluated or the full type rating training shall be accomplished.

Bombardier	Global Express 1-A-10 1-A-11		BD700
British Aerospace Avro	ATP Jetstream 61		BAe/ATP/Jetstream 61
	AVROTO RJ series 146-100 series 200 series 300 series		AVROTORJ/Bae46
British Aerospace Avro	BAC1-11-200 series 400 series 500 series		BAC1-11
Hawker Siddeley/ BAe/ Raytheon	HS125	(D)	HS125
	Bae 125-800 series 1000 series		
Hawker Siddeley/BAe	Jetstream 3100 series 3200 series		Jetstream31/32
BAe/Avro	Jetstream 41		Jetstream41
Canadair (Bombardier)	CL145		CL145
	(Challenger Series) CL 600 CL 601-1A CL 601-3A		CL600/501

	(Challenger) CL604		CL604
	(Regional Jet series) CRJ 100 200	(D)	CRJ100
	700 900		
Casa	C212 series		C212
	CN-235		CN235
Cessna	C 500	(D)	C500/550/560
	C 550 CS 550		
	CS 550 Bravo		
	C 560 Encore		
	C 560XL		C560XL
	C650 Citation III Citation VI Citation VII		C650
	C750 Citation X		C750
Consolidated Vultee Aircraft	CV 240-4	(D)	CV240/340/440
	CV 340 CV 440		
	CV 580		CV580
Dassault	Falcon 10		Falcon 10/100
	Falcon 100		
	Falcon 20 series	(D)	Falcon 20/200
	Falcon 200		
	Falcon 50	(D)	Falcon 50/900
	Falcon 900		
	Falcon 900 EX		
	Falcon 2000	(D)	Falcon 2000/2000 EX
	Falcon 2000 EX (2)		
	Falcon 2000 EX EASy		Falcon 2000EX EASy

(3) The differences training course is valid from the Falcon 2000 to the Falcon 2000EX for crew members previously qualified on the Falcon 2000. The Falcon 2000EX to the Falcon 2000 differences training shall be evaluated or the full type rating training shall be accomplished.

1 Manufacturer	2 Aeroplanes	3	Licence Endorsement
Havilland Canada (Bombardier)	DHC7		DHC7
	DHC8-100 series 200 series 300 series	(D)	DHC8
	DHC8 400 series		
Dornier	DO 328-100		DO 328-100
	DO328-200		DO328-200
	Douglas A-26B		DCA26
	Douglas -3A-51C3G		DC3

McDonnel-Douglas	DC4		DC4
	DC6 series		DC6
	DC7C		DC7
McDonnel-Douglas/Boeing	DC8-33 50,60,70 series		DC8
	DC9 10-50 series		DC9 10-50
	DC9 60 series	(D)	DC9 50/MD58/MD90
	MD 58 series MD 90 series		
	DC10 series		DC10
	MD11		MD11
Embraer	EMB120 Brasillia		EMB120
	EMB 145 135 145 series		EMB 135/145
	EMB 170-100		EMB170
Fokker/Fairchild	FH227 F27/A/F/J F27 series		F27
	F28 series		F28
	F50		F50
	F70 F100		F70/100
Grumman Gulfstream	Grumman G-159		Gulfstream I
	Grumman G1159	(D)	Gulfstream II/III
	Grumman G1159A		
	Gulfstream 1159C/GIV		Gulfstream IV
	Gulfstream V	(D)	G-V
	G550/G500/G V-SP		
Handley Page	Herald series		Herald
Israel Aircraft Industry	IAI-1121 Jetcommander 1123 Commander Jet 1124 Westwind		IAI1121/23/24
	IAI 1124 Astra		IAI1125
Junkers	Junkers 52		JU52
Lockheed	L188 Electra series A	(D)	L188 Electra
	L185 Electra series C		
	L382 G(C 130)		Hercules
	L1011 series		L1011
	L1329		Jetstar
Learjet (Bombardier)	Learjet-20 series	(D)	Learjet20/30
	30 series		
	Learjet-45 series		Learjet45
	Learjet-55 series		Learjet55
	Learjet-60 series		Lerajet60
Leteckee	L410UVP		LetL410
MBB	HFB320		HFB320
	VFW 614		VFW-614
PI Industry	IPTN CN 235-110		IPTNCN235

Rockwell International	NA-265 series		NA265
Saab	SAAB SF340 series		SAAB340
	SAAB 2000		SAAB2000
Short Brothers (Bombardier)	SD3-30	(D)	SD3-30/60
	60		
	Belfast		Belfast
Vickers-Armstrong	Vanguard		Vanguard
	Viscount		Viscount

(4) List of class/type of helicopters – Table 9

This table includes helicopters type certificated under FAR/JAR 27, BCAR and helicopters type certificated by another competent authority acceptable to the Director, but does not include:

(a) helicopters not type certificated in accordance with FAR/JAR 27, BCAR; or

(b) helicopters type certificated in South Africa under special registration such as military, ex-military, experimental (NTCA) or vintage helicopters;

(c) Helicopters not listed in the table may be entered into a South African licence subject to any specific requirements that may be laid down by the Director from time-to-time.

(d) Explanation of the Table:

(i) If a dividing line exists in column 2, this indicates a variant.

(ii) The symbol (D) between variants of types of helicopter used in column 3 indicates that differences training is required.

(iii) Although the licence endorsement (column 4) contains all helicopters listed in column 2, the required familiarisation or differences training has still to be completed (details of differences training can be found in SA-CATS 61.09);

(iv) The specific variant on which the skills test for the type rating has been completed will be recorded according to SA-CATS 61.09.

Table 9

1 Manufacturer	2 Aeroplanes	3	4 Licence Endorsement
Augusta			
SE Turbine	A 119 Koala		A119
ME Turbine	A 109A A 109A II A 109C	(D)	A109/109K/109E
	A 109K		
	A 109E		
Augusta-Bell			
SE Piston	Augusta Bell 47G-2		Bell 47
	Augusta Bell 47G-2A-1		
	Augusta Bell 47G-3B.1		
	Augusta Bell 47G-4		
	Augusta Bell 47G-4A		
	Augusta Bell 47J		
	Augusta Bell 47J-2		
	Augusta Bell 47J-3		
	Augusta Bell 206A Augusta Bell 206B	(D)	Bell206/206L

1 Manufacturer	2 Aeroplanes	3	4 Licence Endorsement
SE Turbine	Augusta Bell 206L		
	Augusta Bell 204	(D)	Bell204/205/UH-1D
	Augusta Bell 205		
ME Turbine	Augusta Bell 212		Bell212/412
	Augusta Bell 412		
	Augusta Bell 412 SP		
Augusta Sikorsky			
ME Turbine	Augusta S-61N1		SK-61
Bell Helicopters			
SE Piston	Bell 47D Bell 47G Bell 47G-1 Bell 47G-2 Bell 47G-3B-1 Bell 47G-4 Bell 47G-4A Bell 47G-5 Bell 47H-1 Bell 47J Bell 47J-2 Bell 47J-2A		Bell47
SE Turbine	Bell 47T		Bell47T
	Bell 47TA		
	Bell 204		Bell204/205/UH-1D
	UH-1D		
	Bell 206 A Bell 206 B Bell 206 B 2 Bell 206 B 3 Bell 206 L Bell 206 L-1 Bell 206 L-3 Bell 206 L-4		Bell206/206L
	Bell 214 B Bell 214 B 1		Bell214
	Bell 407		Bell407
ME Turbine	Bell 206 LT Twinranger		Bell 206LT
	Bell 212		Bell212/412
	Bell 412 SP Bell 412 HP Bell 412 EP	(D)	
	Bell 214 ST		Bell214ST
	Bell 222 Bell 222 A Bell 222 B Bell 222 UT Bell 222 SP	(D)	Bell222/230/430

1 Manufacturer	2 Aeroplanes	3	4 Licence Endorsement
	Bell 230		
	Bell 430		
	Bell 427		Bell427
Boeing-Vertol			
ME Turbine	Boeing 234 LR		BV234
Bristol Aircraft			
SE Piston	B-171-B		Bristol 171B
Brantly			
SE Piston	B-2 B-2 B		Brantly B2
Breda Nardi			
SE Piston	Breda Nardi 269		HU269
SE Turbine	Breda Nardi 369		HU369
EH Industries			
ME Turbine	EH101		EH101
Enstrom			
SE Piston	F28 A-D F28 C 2 F280C F280F F280FX F280D		EN28
SE Turbine	F480		ENF480
Eurocopter			
SE Turbine	AS350B AS350B 1 AS350B 2 AS350D AS350B A AS350B B	(D)	AS350/350B3
	AS350B 3		
	EC130B 4		EC130B4
	EC120		EC120
	SA341 G SA342 J		SA341/342
	SA3180 SA318 B SA318 C SE3130 SE313 B		SA318/SE313
	SE3160 SE316 B SE316 C	(D)	SA316/319/315
	SA319 B		
	SA315 B		
	SA360		SA360

1 Manufacturer	2 Aeroplanes	3	4 Licence Endorsement
	SO1221		SO1221
ME Turbine	AS332 C AS332 C 1 AS332 L AS332 L 1	(D)	AS332/332L2
	AS332 L 2		
	EC 225 L P		EC 225LP
	AS355 E AS355 F AS355 F 1 AS355 F 2	(D)	AS355/355N
	AS355 N		
	BO105 A BO105 C BO105 D BO105 LS A-1 BO105 LS A-3 BO105 S BO105 CBS		BO105/105LS/105CBS
	EC135 T1 CDS EC135 P1 CDS	(D)	EC135
	EC135 T1 CPDS EC135 P1 CPDS EC135 T2 CPDS EC135 P2 CPDS		
	MBB-BK 117 A-1 MBB-BK 117 A-3 MBB-BK 117 A-4 MBB-BK 117 B-1 MBB-BK 117 B-2	(D)	BK117
	MBB-BK 117 C-1		
	MBB-BK 117 C-2		
	SA330 F SA330 G SA330 J		SA330
	SA365 SA365 C 1 SA365 C 2 SA365 C 3	(D)	SA365/365N
	SA365 N SA365 N 1 SA365 N 2		
	EC155		EC155
Hiller			
SE Piston	UH12 A UH12 B UH12 E		UH12

1 Manufacturer	2 Aeroplanes	3	4 Licence Endorsement
Hughes/Schweitzer			
SE Piston	269 A 269 B 269 C 300 C 300 CB		HU269
SE Turbine	330SP		SC330
Kaman			
SE Turbine	Kaman K 1200		K1200
McDonnell Douglas Helicopters			
SE Turbine	Hughes 369 D Hughes 369 E Hughes 369 HE Hughes 369HS MD500 N (NOTAR) MD520 N	(D)	HU369/MD500N/ 600
	MD600		
ME Turbine	MD900	(D)	MD900/902
	MD902		
Robinson			
SE Piston	R22 R22 A R22 B		R22
	R44		R44
Silvercraft			
SE Piston	SV 4		SV4
Sikorsky			
SE Piston	S55		SK55
	S58		SK58
ME Piston	S58T		SK58T
	S76 A	(D)	SK76/76B/76C/76C+
	S76 A+		
	S76 A++		
	S76 B		
	S76 C		
	S76 C+		
	SN-61 N		SK61
	SN-61 S		
Westland			
SE Piston	Westland Bell 47 G3 B-1		Bell 47
Westland Helicopters			
SE Piston	Westland S55 Series 1	(D)	WHS55
SE Turbine	Westland S55 Series 3		
Ministry of Aviation Industry of Russia			
ME Piston	Kamov KA26 D		KA26D
ME Turbine	KA32 A		KA32

1 Manufacturer	2 Aeroplanes	3	4 Licence Endorsement
	MIL Mi-8 MIL Mi 17 MIL Mi 171 MIL Mi 172		Mi8
P.Z.L Swidnik, Poland			
ME Turbine	MIL Mi-2		Mi-2
	PZL Kania		KANIA
	PZL W-3	(D)	W-3SOKOL
	PZL W-34		

Table 10 – Warbird Type Rating Standards
Standards for the issue of a Warbird Type Rating for the following aircraft are detailed below –

1 Manufacturer	2 Aeroplanes	3	4 Licence Endorsement
Boomerang			
	CAC CA-12 (all models)		
	CAC CA-13 (all models)		
	CAC CA-19 (all models)		
	A 109E		
Canberra			
	Handley Page Canberra TT18		
	English Electric Canberra (all models)		
Cessna			
	Cessna A37 Dragonfly (all models)		Cessna A37
L-39			
	Aero Vodochnyody L-39 Albatross (all models)		
L-29			
	Aero Vodochnyody L-29 Delphin (all models)		
Firefly			
	Fairey Firefly (all models)		
Fiat G59			
	Fiat G59 (all models)		
Fury			
	Hawker Sea Fury (all models) Hawker Fury (all models)		
Hunter			
	Hawker Hunter (all models)		
Iskra			
	PZL TS-11 Iskra		

1 Manufacturer	2 Aeroplanes	3	4 Licence Endorsement
Kittyhawk			
	Curtis P-40 (all models)		
Hudson			
	Lockheed 414 Hudson (all models)		
Ventura			
	Lockheed PV-1 (all models)		
ME109			
	Messerschmit BF 109 (all models)		
Meteor			
	Gloster Meteor (all models)		
MIG 15			
	Mikoyan MIG-15 (all models)		
MIG 17			
	Mikoyan MIG-17 (all models)		
MIG 21			
	Mikoyan MIG-21 (all models)		
Mustang			
	CAC CA-17 (all models) CAC CA-18 (all models) North American P-51 (all models)		
Sabre			
	CAC CA-27 (all models)		
North American			
	F-86 (all models)		
Spitfire			
	Supermarine Spitfire (all models) Supermarine Seafire (all models)		
Strikemaster			
	BAC-167 (all models) BAC Jet Provost (all models)		
Trojan			
	North American T28 (all models)		
Avenger			
	Grumman TBM Avenger (all models)		
Beaufort			
	Bristol Beaufort (all models)		
Vampire			
	De Havilland DH-115 Vampire (all models)		

1 Manufacturer	2 Aeroplanes	3	4 Licence Endorsement
YAK 3			
	Yakovlev YAK 3 (all models)		
All ex-military aircraft			

61.10.1 REQUIREMENTS FOR A NIGHT RATING

1. Theoretical Knowledge Instruction

(1) Night rating: General

The aim of the night rating theoretical knowledge instruction syllabus referred to in CAR 61.10.1(2)(a) is to ensure that the applicant has a thorough understanding of the theoretical aspects surrounding the night rating. Night flying takes place in a potentially hostile environment and applicants must understand each element of the environment in which they are operating.

(a) Air Law –

(i) The definition of night flying;
(ii) The privileges and limitations associated with the night rating;
(iii) The pilot-in-command's responsibilities;
(iv) The equipment to be carried on board for night flying;
(v) Aircraft lighting including navigation lights;
(vi) VFR differences from day flying;
(vii) Aerodrome requirements for night flying.

(b) Meteorology –

(i) The formation of fog;
(ii) Various types of fog;
(iii) Katabatic winds;
(iv) Mixing, veering and backing of winds at night;
(v) Formation of ice and frost;
(vi) Nocturnal Thunderstorms.

(c) Human performance –

(i) Factors affecting night vision; the preservation of night vision;
(ii) Visual illusions;
(iii) Hypoxia;
(iv) Vertigo;
(v) Autokinesis.

(d) Lighting systems –

(i) External aircraft lighting;
(ii) Internal cockpit lighting;
(iii) Taxiway lighting;
(iv) Runway lighting;
(v) Approach lighting systems;
(vi) Obstruction lighting;
(vii) Aerodrome identification beacons;
(viii) Where to find information on lighting systems;
(ix) Pilot-operated lighting.

2. Training

(1) Contents and requirements of training course

The applicant must complete a training course with an aviation training organisation approved by the Director in terms of Part 141. However, the theoretical knowledge course

and the practical training course may be completed at different aviation training organisations.

3. Practical Instruction

For the detailed practical training syllabus, refer to Appendix 11.0 to Document SA-CATS 61.

(1) Helicopter

In the case of an applicant for a night rating for a helicopter the practical instruction must include –

(a) Pre-flight Operations;
(b) Take-off Procedures;
(c) In-flight Manoeuvres;
(d) Approach and Landing; and
(e) Non-normal Emergency Operations

(2) Aeroplane

In the case of an applicant for a night rating for an aeroplane the practical instruction must include –

(a) Pre-flight Operations;
(b) Take-off Procedures;
(c) In-flight Manoeuvres;
(d) Approach and Landing; and
(e) Non-normal Emergency Operations.

61.11.1 REQUIREMENTS FOR AN INSTRUMENT RATING

1. Training

(1) Aim of the Instrument Rating training course

The aim of the Instrument Rating training course is to train a candidate to the level of proficiency required for the issue of an instrument rating.

(2) Contents and requirements of training course

(a) The applicant must complete a training course with an aviation training organisation approved by the Director in terms of Part 141. However, the theoretical knowledge course and the practical training course may be completed at different aviation training organisations.

(b) The course is comprised of –

(i) theoretical knowledge course; and
(ii) practical training course.

(3) Theoretical knowledge course

(a) An instrument rating theoretical knowledge course shall cover the subjects listed below –

(b) IF theoretical knowledge –

(i) Aviation Meteorology;
(ii) Flight Planning and Performance;
(iii) Radio Aids and Communication;
(iv) General Navigation;
(v) Air Law and Operational Procedures;
(vi) Instruments and Electronics;
(vii) Human Performance;

(c) Theoretical knowledge course syllabus

The detailed theoretical knowledge syllabus is contained in Appendix 2.0A to Document SA-CATS 61.

(4) Practical Flight Training course

The detailed practical flight training syllabus is contained in Appendix 12.0.

(5) Radio Telephony

(a) To be eligible for the RTC a candidate must prove his knowledge of the ITU (International Telecommunications Union), and CAA requirements in both written and oral tests. The requirements are:

(i) ITU: Knowledge of radiotelephony operation and procedures, ability to transmit and receive messages by radio and knowledge of the radio communication regulations (e.g. use of different appropriate frequencies, interference, etc.) and especially those relating to the safety of human life.

(ii) ICASA: Proficiency in operating a radio installation including changing frequency, transmitting and receiving messages using the prescribed procedures, identification of Morse code beacons and clearing minor external faults.

(iii) The CAA: requirements are based on documents issued by the International Civil Aviation Organisation (ICAO) dealing with procedures, abbreviations and flight planning, the various aeronautical information circulars (AIC) in force, the S.A. Aeronautical Information Publication (AIP), Civil Aviation Regulations (CAR) and Civil Aviation Technical Standards (CATS) and also other relevant regulations or rules in force.

(iv) Applicants for a General Radio Certificate shall pass a theoretical General Radio Examination at an approved CAA examination centre. The training syllabus for a General Radio Certificate is contained in Appendix 1.5a to Document SA-CATS 61.

(b) Applicants for a General Radio Certificate shall in addition to the theoretical knowledge examination pass a practical communication test including the identification of radio beacons using morse code conducted by a Designated Radio Examiner (General). The practical test shall include the full completion of an ATC IFR Flight Plan and examine, at minimum, the following aspects for an IFR flight departing on a standard or non-standard departure from a controlled aerodrome, into a CTR/TMA, flying in both controlled and uncontrolled airspace en route to a controlled airfield as destination, complying with a standard arrival (STAR) and the completion of an instrument approach procedure (precision or non-precision):

(i) Use of Radio on the Ground –

(aa) Obtaining start clearance

(bb) Obtaining taxi clearance

(cc) Acceptance and read back of ATC departure clearance (Standard/non-standard)

(ii) Departure procedure –

(aa) Take-off clearance

(bb) Use of SID chart/compliance with non-standard departure procedure

(cc) Selection of departure frequency and contact with relevant ATSU

(dd) Use of area chart if applicable

(iii) En route procedures –

(aa) Use of radio navigation chart

(bb) Selection of frequencies appropriate to the route

(cc) Passing and revising estimates

(dd) Complying with onward clearance time (OCT)

(iv) Arrival procedures –

(aa) Use of area chart if applicable

(bb) Acceptance and review of STAR and instrument approach charts

(cc) Radar vectors to ILS localiser or Holding beacon for non-precision approach including expected approach time (EAT)

(dd) Establishing weather minima at landing aerodrome and compliance with Approach Ban

(ee) Completion of instrument approach procedure using IAC and correct radio procedures

(v) Radio communication failure –

Emphasis shall be placed on correct use of aviation words and phrases as well as the avoidance of slang terms.

61.11.3 THEORETICAL KNOWLEDGE EXAMINATION

1. Written examination

(1) The applicant for an instrument rating shall pass the written theoretical knowledge examinations in the subjects prescribed in TS 61.11.1(3)(b).

(2) The written theoretical knowledge examination shall be conducted by the SACAA.

61.11.4 SKILLS TEST FOR AN INSTRUMENT RATING

1. Procedures and manoeuvres

The Skills Test shall be conducted in accordance with the Practical Test Standard contained in Appendix 2.5 to Document SA-CATS 61, as appropriate, using the skills test form.

61.11.7 REVALIDATION OF AN INSTRUMENT RATING

1. Revalidation check

The revalidation check shall be conducted in accordance with the Practical Test Standard contained in Appendix 2.5 to Document SA-CATS 61, as appropriate, using the skills test form.

2. Theoretical knowledge examination

The theoretical knowledge examinations shall be the examinations indicated in TS 61.11.3.

3. Application for revalidation

The application for an instrument rating shall be made on the appropriate form.

4. Endorsement of logbook

The endorsement in the logbook of the applicant shall contain the following –

(a) the stamp of the Designated Flight Examiner, which shall indicate the name, licence number and designation of the Designated Flight Examiner;

(b) the date of the proficiency test;

(c) the description of the proficiency test;

(d) the result of the test as reflected on the test form; and

(e) the signature of the Designated Flight Examiner.

61.12.1 REQUIREMENTS FOR GRADE III AEROPLANE FLIGHT INSTRUCTOR RATING

1. Training

(1) Aim

The aim of the Grade III Flight Instructor (A) rating course is to train a candidate to the level of proficiency required for the issue of a Grade III flight instructor rating.

(2) Contents and requirements of training course

(a) The applicant must complete a training course with an aviation training organisation approved by the Director in terms of Part 141. However, the theoretical knowledge course and the practical training course may be completed at different aviation training organisations.

(b) A SA Air Force pilot or navigator instructor may be exempted from attending the theoretical knowledge course. The Air Force pilot instructor is furthermore exempted from the requirement to conduct 20 hours of patter.

(3) Theoretical knowledge course

Theoretical knowledge syllabus is contained in Appendix 13.0 to Document SA-CATS 61.

(4) Practical Instruction course

The detailed syllabus is contained in Appendix 13.1 to Document SA-CATS 61.

(5) Ground evaluation

The ground evaluation shall comprise the following:

(a) a class teaching evaluation test conducted by a Designated Flight Examiner and the instructor responsible for the training of the candidate;

(b) the DFE conducting such test shall be specifically nominated for FIC by the Director;

(c) the candidate shall present a full briefing on a subject that was given in advance to him or her by the Designated Flight Examiner;

(d) the candidate shall be assessed on their basic instrument training aspects.

61.12.3 THEORETICAL KNOWLEDGE EXAMINATIONS FOR GRADE III AEROPLANE FLIGHT INSTRUCTOR RATING

1. Syllabus

The applicant for a Grade III flight instructor (A) rating shall pass prior to the commencement of practical class and flight training the written theoretical knowledge examinations in the subjects listed below –

(a) Applied Meteorology and Navigation;

(b) Principles of Flight and Legislation.

61.12.4 SKILLS TEST FOR GRADE III (AEROPLANE) FLIGHT INSTRUCTOR RATING

1. Procedures and manoeuvres

The skills test shall be conducted by the nominated Designated Flight Examiner in accordance with the Practical Test Standard contained in skills test form.

61.12.6 PERIOD OF VALIDITY AND REISSUE OF A GRADE III AEROPLANE FLIGHT INSTRUCTOR RATING

1. Flight Instructor refresher seminar

(1) Flight Instructor refresher seminars should be arranged by ATOs as part of their training programme. Attendance by a flight instructor of the bi-annual instructor seminar conducted regionally by the Authority will substitute this requirement.

(2) The seminars should run for between one and two days, and if credit is required in terms of this Part, attendance from participants will be required for the whole duration of the seminar including breakout groups or workshops.

(3) The Director will make use of the services of experienced flight instructors and Designated Flight Examiners who are well versed in various levels of flying training to participate in, or to provide lectures and group discussions at the seminars.

(4) The attendance form shall be completed and signed by the organiser of the Seminar and shall accompany the revalidation application.

(5) The content of the Flight Instructor refresher seminar will generally be selected from the following topics –

(a) new and/or current rules/regulations, with emphasis on knowledge of SA CAR as applicable to the job of the flight instructor;

(b) teaching and learning;

(c) instructional techniques;

(d) the role of the instructor;

(e) human factors;

(f) topical and recent accidents and their probable cause;

(g) flight safety, incident and accident prevention;

(h) airmanship;

(i) legal aspects and enforcement procedures;

(j) navigational skills including new/current radio navigation aids;

(k) teaching instrument flying;

(l) weather related topics including methods of distribution of aeronautical information; and

(m) feedback on knowledge and skills deficiencies revealed in the prescribed theoretical and practical examinations and tests, for improvement in instruction.

(6) The seminar will consist of formal sessions, which will typically allow for a presentation time of 45 minutes and 15 minutes for questions. Group breakouts and discussions will be facilitated and summarised at the end of the day's proceedings.

61.12.7 REVALIDATION OF A GRADE III AEROPLANE FLIGHT INSTRUCTOR RATING

1. Flight instructor refresher seminar

(a) The Flight Instructor refresher seminar is described in TS 61.12.6.

2. Revalidation skills test

(a) The revalidation skills test shall be conducted by the nominated Designated Flight Examiner in accordance with the Practical Test Standard contained in skills test form.

61.13.1 REQUIREMENTS FOR GRADE II AEROPLANE FLIGHT INSTRUCTOR RATING

1. Training

The aim of the Grade II Flight Instructor (A) rating course is to train a candidate to the level of proficiency required for the issue of a Grade II flight instructor rating.

2. Contents and requirements of training course

(1) The applicant must complete a training course with an aviation training organisation approved by the Director in terms of Part 141. However, the theoretical knowledge course and the practical training course may be completed at different aviation training organisations.

(2) A SA Air Force pilot or navigator instructor may be exempted from attending the theoretical knowledge course. The Air Force pilot instructor is furthermore exempted from the requirement to conduct 20 hours of patter.

3. Theoretical knowledge course

Theoretical knowledge syllabus is contained in Appendix 13.0 to Document SA-CATS 61.

4. Practical instruction course

The detailed syllabus is contained in Appendix 13.1 to Document SA-CATS 61.

5. Ground evaluation

The ground evaluation shall comprise the following:

(a) a class teaching evaluation test conducted by a Designated Flight Examiner and the instructor responsible for the training of the candidate;

(b) the DFE conducting such test shall be nominated by the Director;

(c) the candidate shall present a full briefing on a subject given to him or her in advance by the nominated Designated Flight Examiner.

61.13.4 SKILLS TEST FOR GRADE II AEROPLANE FLIGHT INSTRUCTOR RATING

1. Procedures and manoeuvres

The Skills Test shall be conducted by the nominated Designated Flight Examiner in accordance with the Practical Test Standard contained in skills test form.

61.13.5 PRIVILEGES AND LIMITATIONS OF A GRADE II AEROPLANE FLIGHT INSTRUCTOR RATING

1. Endorsement

The flight instructor wishing to conduct training for the ratings listed in CAR 61.13.5(1)(g) shall have the applicable endorsement entered into his or her pilot logbook.

61.13.6 PERIOD OF VALIDITY AND REISSUE OF A GRADE II AEROPLANE FLIGHT INSTRUCTOR RATING

1. Flight Instructor refresher seminar

(1) Flight Instructor refresher seminars will be co-ordinated and arranged by the SA CAA at various centres, taking due regard of the distribution of instructors in South Africa.

(2) The seminars will run for between one and two days, and if credit is required in terms of this Part, attendance from participants will be required for the whole duration of the seminar including breakout groups/workshops.

(3) The Director will make use of the services of experienced flight instructors and Designated Flight Examiners who are well versed in various levels of flying training to participate in, or to provide lectures and group discussions at the seminars.

(4) The attendance form shall be completed and signed by the organiser of the Seminar and shall accompany the revalidation application.

(5) The content of the Flight Instructor refresher seminar will generally be selected from the following topics –

(a) new and/or current rules/regulations, with emphasis on knowledge of SA CAR as applicable to the job of the flight instructor.

(b) teaching and learning;

(c) instructional techniques;

(d) the role of the instructor;

(e) human factors;

(f) topical and recent accidents and their probable cause;

(g) flight safety, incident and accident prevention;

(h) airmanship;

(i) legal aspects and enforcement procedures;

(j) navigational skills including new/current radio navigation aids;

(k) teaching instrument flying;

(l) weather related topics including methods of distribution of aeronautical information; and

(m) feedback on knowledge and skills deficiencies revealed in the prescribed theoretical and practical examinations and tests, for improvement in instruction.

(6) The seminar will consist of formal sessions, which will typically allow for a presentation time of 45 minutes and 15 minutes for questions. Group breakouts and discussions will be facilitated and summarised at the end of the day's proceedings.

61.13.7 REVALIDATION OF A GRADE II AEROPLANE FLIGHT INSTRUCTOR RATING

1. Flight instructor refresher seminar

(1) The Flight Instructor refresher seminar is described in TS 61.13.6.

2. Revalidation skills test

(1) The revalidation skills test shall be conducted by the nominated Designated Flight Examiner in accordance with the Practical Test Standard contained in skills test form.

61.14.1 REQUIREMENTS FOR GRADE I AEROPLANE FLIGHT INSTRUCTOR RATING

1. Training course

(1) The applicant must complete a training course with a training organisation approved by the Director. The detailed syllabus is contained in Appendix 18.0 to Document SA-CATS 61.

(2) The applicant must pass an oral examination on the subjects contained in Appendix 18.0 to Document SA-CATS 61.

61.14.4 SKILLS TEST FOR GRADE I AEROPLANE FLIGHT INSTRUCTOR RATING

1. Procedures and manoeuvres

The skills test shall be conducted in accordance with the Practical Test Standard contained in the skills test form.

61.14.7 REVALIDATION OF GRADE I AEROPLANE FLIGHT INSTRUCTOR RATING

1. Flight instructor refresher seminar

(1) The skills test shall be conducted in accordance with the Practical Test Standard contained in the skills test form. This form, completed by the Designated Flight Examiner, shall accompany the application form.

(2) When required by CAR 61.18.6, the applicant for a revalidation of a Grade III flight instructor (A) rating shall complete the flight instructor refresher seminar as detailed below –

(a) Flight Instructor refresher seminar

(i) Flight Instructor refresher seminars will be co-ordinated and arranged by the SA CAA at various centres, taking due regard of the distribution of instructors in South Africa.

(ii) The seminars will run for between one and two days, and if credit is required in terms of this Part, attendance from participants will be required for the whole duration of the seminar including breakout groups/workshops.

(iii) The Director will make use of the services of experienced flight instructors and Designated Flight Examiners who are well versed in various levels of flying training to participate in, or to provide lectures and group discussions at the seminars.

(iv) The attendance form shall be completed and signed by the organiser of the Seminar and shall accompany the revalidation application.

(v) The content of the Flight Instructor refresher seminar will generally be selected from the following topics –

(aa) new and/or current rules/regulations, with emphasis on knowledge of SA CAR's as applicable to the job of the flight instructor;

(bb) teaching and learning;

(cc) instructional techniques;

(dd) the role of the instructor;

(ee) human factors;

(ff) topical and recent accidents and their probable cause;

(gg) flight safety, incident and accident prevention;

(hh) airmanship;

(ii) legal aspects and enforcement procedures;

(jj) navigational skills including new/current radio navigation aids;

(kk) teaching instrument flying;

(ll) weather related topics including methods of distribution of aeronautical information; and

(mm) feedback on knowledge and skills deficiencies revealed in the prescribed theoretical and practical examinations and tests, for improvement in instruction.

(vi) The seminar will consist of formal sessions, which will typically allow for a presentation time of 45 minutes and 15 minutes for questions. Group breakouts and discussions will be facilitated and summarised at the end of the day's proceedings.

(3) The Skills Test report referred to is the skills test form.

(4) The result of the skills test shall be endorsed in the pilot logbook as per CAR 61.01.19(2) and Appendix A to this Document.

61.15.1 REQUIREMENTS FOR GRADE III HELICOPTER FLIGHT INSTRUCTOR RATING

1. Training

The aim of the Grade III Flight Instructor (H) rating course is to train a candidate to the level of proficiency required for the issue of a Grade III flight instructor rating.

2. Contents and requirements of training course

(1) The applicant must complete a training course with an aviation training organisation approved by the Director in terms of Part 141. However, the theoretical knowledge course

and the practical training course may be completed at different aviation training organisations.

(2) A SA Air Force pilot or navigator instructor is exempted from attending the theoretical knowledge course. The Air Force pilot instructor is furthermore exempted from the requirement to conduct 20 hours of patter.

3. Theoretical knowledge course

Theoretical knowledge syllabus is contained in Appendix 13.0 to Document SA-CATS 61.

4. Practical instruction course

The detailed syllabus is contained in Appendix 13.1 to Document SA-CATS 61.

5. Theoretical knowledge examinations

The applicant for a Grade III flight instructor (H) rating shall pass prior to the commencement of training the written theoretical knowledge examinations in the subjects listed below –

(a) Applied Meteorology and Navigation;

(b) Principles of Flight and Legislation.

61.15.3 THEORETICAL KNOWLEDGE EXAMINATIONS FOR GRADE III HELICOPTER FLIGHT INSTRUCTOR RATING

1. Syllabus

The applicant for a Grade III flight instructor (H) rating shall pass prior to the commencement of practical class and flight training the written theoretical knowledge examinations in the subjects listed below –

(a) Applied Meteorology and Navigation;

(b) Principles of Flight and Legislation.

61.15.4 SKILLS TEST FOR A GRADE III HELICOPTER FLIGHT INSTRUCTOR RATING

1. Procedures and manoeuvres

(1) The DFE conducting the skills test shall be nominated by the Director.

(2) The skills test shall be conducted by the nominated Designated Flight Examiner in accordance with the Practical Test Standard contained in skills test Form.

(3) The skills test shall comprise a class teaching evaluation test conducted by a Designated Flight Examiner and the instructor responsible for the training of the candidate.

(4) The candidate shall present a full briefing on a subject given to him or her in advance by the nominated Designated Flight Examiner.

61.15.6 PERIOD OF VALIDITY AND REISSUE OF A GRADE III HELICOPTER FLIGHT INSTRUCTOR RATING

1. Flight Instructor refresher seminar

The flight instructor refresher seminar shall be in the format as prescribed in TS 61.15.7.

61.15.7 REVALIDATION OF A GRADE III HELICOPTER FLIGHT INSTRUCTOR RATING

1. Flight Instructor refresher seminar

(1) Flight Instructor refresher seminars will be co-ordinated and arranged by the SA CAA at various centres, taking due regard of the distribution of instructors in South Africa.

(2) The seminars will run for between one and two days, and if credit is required in terms of this Part, attendance from participants will be required for the whole duration of the seminar including breakout groups/workshops.

(3) The Director will make use of the services of experienced flight instructors and Designated Flight Examiners who are well versed in various levels of flying training to participate in, or to provide lectures and group discussions at the seminars.

(4) The attendance form shall be completed and signed by the organiser of the Seminar and shall accompany the revalidation application.

(5) The content of the Flight Instructor refresher seminar will generally be selected from the following topics –

(a) new and/or current rules/regulations, with emphasis on knowledge of SA CAR as applicable to the job of the flight instructor;
(b) teaching and learning;
(c) instructional techniques;
(d) the role of the instructor;
(e) human factors;
(f) topical and recent accidents and their probable cause;
(g) flight safety, incident and accident prevention;
(h) airmanship;
(i) legal aspects and enforcement procedures;
(j) navigational skills including new/current radio navigation aids;
(k) teaching instrument flying;
(l) weather related topics including methods of distribution of aeronautical information; and
(m) feedback on knowledge and skills deficiencies revealed in the prescribed theoretical and practical examinations and tests, for improvement in instruction.

(6) The seminar will consist of formal sessions, which will typically allow for a presentation time of 45 minutes and 15 minutes for questions. Group breakouts and discussions will be facilitated and summarised at the end of the day's proceedings.

2. Revalidation skills test

(2) The skills test shall be conducted in accordance with the Practical Test Standard contained in skills test form CA 61.16.3.

(3) The Grade III instructor rating revalidation shall be endorsed in the applicant's logbook as prescribed in Appendix A to Document SA-CATS 61.

(4) The failure of the Grade III instructor rating revalidation shall be endorsed in the applicant's logbook as prescribed in Appendix A to Document SA-CATS 61.

61.16.1 REQUIREMENTS FOR GRADE II HELICOPTER FLIGHT INSTRUCTOR RATING

1. Training

The aim of the Grade II Flight Instructor (H) rating course is to train a candidate to the level of proficiency required for the issue of a Grade II helicopter flight instructor rating.

2. Contents and requirements of training course

(1) The applicant must complete a training course with an aviation training organisation approved by the Director in terms of Part 141. However, the theoretical knowledge course and the practical training course may be completed at different aviation training organisations.

(2) A SA Air Force pilot or navigator instructor may be exempted from attending the theoretical knowledge course. The Air Force pilot instructor is furthermore exempted from the requirement to conduct 20 hours of patter.

3. Theoretical knowledge course

Theoretical knowledge syllabus is contained in Appendix 13.0 to Document SA-CATS 61.

4. Practical instruction course

The detailed syllabus is contained in Appendix 13.1 to Document SA-CATS 61.

5. Ground evaluation

(1) The ground evaluation shall comprise a class teaching evaluation test conducted by a Designated Flight Examiner and the instructor responsible for the training of the candidate.

(2) The DFE conducting the test shall be nominated by the Director.

(3) The candidate shall present a full briefing on a subject given to him or her in advance by the nominated Designated Flight Examiner.

61.16.4 SKILLS TEST FOR GRADE II HELICOPTER FLIGHT INSTRUCTOR RATING

1. Procedures and manoeuvres

The Skills Test shall be conducted by the nominated Designated Flight Examiner in accordance with the Practical Test Standard contained in skills test Form CA 61-16-3.

61.16.5 PRIVILEGES AND LIMITATIONS OF A GRADE II HELICOPTER FLIGHT INSTRUCTOR RATING

1. Endorsement

The flight instructor wishing to conduct training for the ratings listed in subparagraph (1)(h) of regulation 61.16.5 shall have the applicable endorsement entered into his or her pilot logbook.

61.16.6 PERIOD OF VALIDITY AND REISSUE OF A GRADE II HELICOPTER FLIGHT INSTRUCTOR RATING

1. Flight Instructor refresher seminar

(1) Flight Instructor refresher seminars will be co-ordinated and arranged by the SA CAA at various centres, taking due regard of the distribution of instructors in South Africa.

(2) The seminars will run for between one and two days, and if credit is required in terms of this Part, attendance from participants will be required for the whole duration of the seminar including breakout groups/workshops.

(3) The Director will make use of the services of experienced flight instructors and Designated Flight Examiners who are well versed in various levels of flying training to participate in, or to provide lectures and group discussions at the seminars.

(4) The attendance form shall be completed and signed by the organiser of the Seminar and shall accompany the revalidation application.

(5) The content of the Flight Instructor refresher seminar will generally be selected from the following topics –

(a) new and/or current rules/regulations, with emphasis on knowledge of SA CAR as applicable to the job of the flight instructor;
(b) teaching and learning;
(c) instructional techniques;
(d) the role of the instructor;
(e) human factors;
(f) topical and recent accidents and their probable cause;
(g) flight safety, incident and accident prevention;
(h) airmanship;
(i) legal aspects and enforcement procedures;
(j) navigational skills including new/current radio navigation aids;
(k) teaching instrument flying;
(l) weather related topics including methods of distribution of aeronautical information; and
(m) feedback on knowledge and skills deficiencies revealed in the prescribed theoretical and practical examinations and tests, for improvement in instruction.

(6) The seminar will consist of formal sessions, which will typically allow for a presentation time of 45 minutes and 15 minutes for questions. Group breakouts and discussions will be facilitated and summarised at the end of the day's proceedings.

61.16.7 REVALIDATION OF A GRADE II HELICOPTER FLIGHT INSTRUCTOR RATING

1. Flight instructor refresher seminar

(1) The Flight Instructor refresher seminar is described in TS 61.16.6.

2. Revalidation skills test

(1) The revalidation skills test shall be conducted by the nominated Designated Flight Examiner in accordance with the Practical Test Standard contained in skills test form CA 61.16.3.

61.17.1 REQUIREMENTS FOR GRADE I HELICOPTER FLIGHT INSTRUCTOR RATING

1. Training course

(1) The applicant must complete a training course with a training organisation approved by the Director. The detailed syllabus is contained in Appendix 18.0 to Document SA-CATS 61.

(2) The applicant must pass an oral examination on the subjects contained in Appendix 18.0 to Document SA-CATS 61.

61.17.4 SKILLS TEST FOR GRADE I HELICOPTER FLIGHT INSTRUCTOR RATING

1. Procedures and manoeuvres

The Skills Test shall be conducted in accordance with the Practical Test Standard contained in the skills test form CA 61-16-3.

61.17.7 REVALIDATION OF GRADE I HELICOPTER FLIGHT INSTRUCTOR RATING

1. General

(1) The skills test shall be conducted in accordance with the Practical Test Standard contained in the skills test Form CA 61-16-3. This form, completed by the Designated Flight Examiner, shall accompany the application form.

(2) When required by sub-regulation 61.17.6 (3), the applicant for a revalidation of a Grade III flight instructor (H) rating shall complete the flight instructor refresher seminar as detailed below.

2. Flight Instructor refresher seminar

(1) Flight Instructor refresher seminars will be co-ordinated and arranged by the SA CAA at various centres, taking due regard of the distribution of instructors in South Africa.

(2) The seminars will run for between one and two days, and if credit is required in terms of this Part, attendance from participants will be required for the whole duration of the seminar including breakout groups/workshops.

(3) The Director will make use of the services of experienced flight instructors and Designated Flight Examiners who are well versed in various levels of flying training to participate in, or to provide lectures and group discussions at the seminars.

(4) The attendance form shall be completed and signed by the organiser of the Seminar and shall accompany the revalidation application.

(5) The content of the Flight Instructor refresher seminar will generally be selected from the following topics –

(a) new and/or current rules/regulations, with emphasis on knowledge of SA CAR's as applicable to the job of the flight instructor;
(b) teaching and learning;
(c) instructional techniques;
(d) the role of the instructor;
(e) human factors;
(f) topical and recent accidents and their probable cause;
(g) flight safety, incident and accident prevention;
(h) airmanship;
(i) legal aspects and enforcement procedures;
(j) navigational skills including new/current radio navigation aids;
(k) teaching instrument flying;
(l) weather related topics including methods of distribution of aeronautical information; and
(m) feedback on knowledge and skills deficiencies revealed in the prescribed theoretical and practical examinations and tests, for improvement in instruction.

(6) The seminar will consist of formal sessions, which will typically allow for a presentation time of 45 minutes and 15 minutes for questions. Group breakouts and discussions will be facilitated and summarised at the end of the day's proceedings.

3. Skills test

(1) The skills test report referred to is the skills test Form CA 61-16-3.

(2) The result of the skills test shall be endorsed in the pilot logbook as per subregulation 61.1.19.(2) and Appendix A to SA-CATS-FCL 61.

61.22.1 REQUIREMENTS FOR HELICOPTER SLING LOAD RATING

1. Training

(1) Aim of training course

The aim of the training course is to train a candidate to the level of proficiency required for the issue of a helicopter sling load rating, and to provide the training necessary to act as pilot-in-command of a helicopter engaged in sling load operations.

(2). Contents and requirements of training course

(a) The candidate must have completed not less than 250 hours of flight time as pilot-in-command of a helicopter. The course must be conducted by the holder of an aviation training organisation approval, issued by the Director in terms of Part 141.

(b) The course comprises –

(i) a theoretical knowledge course; and

(ii) a practical training course.

(3) Theoretical knowledge course

The theoretical knowledge course must comprise instruction on the following –

(a) the significance of operations within and outside ground effect, and the correct use of the relevant performance charts;

(b) the possible fore and aft C of G changes when picking up and releasing sling loads;

(c) the pre-flight checking and correct operation of the helicopter cargo hook equipment, including the emergency release;

(d) the importance of a full and correct briefing for all flight and ground crew members participating in the operation as regards to –

(i) pick-up and drop-off points;

(ii) load preparation and flight characteristics of different loads;

(ii) oscillation characteristics and their control;

(e) the care, selection, preparation and correct use of lifting equipment, including strops of various lengths, swivels, shackles, nets, and safety harnesses for cabin crew, as applicable;

(f) responsibilities and duties of cabin crew;

(g) aircraft-generated static, use of the static discharge pole and the correct procedure in this regard;

(h) marshalling signals;

(i) correct radio procedures and terminology for intercom communications between the pilot and cabin crew;

(j) pick-up and release procedures;

(k) safety and other equipment, including hand-held transceivers, hard hats, safety goggles, durable gloves, overalls and whistles;

(l) emergency procedures, including engine failure in the hover, strops getting fouled either with the helicopter or with other items, loads becoming difficult or impossible to control in flight, and jettisoning of loads; the effects of buildings and obstruction on prevailing winds, escape routes in the event of downdrafts, turbulence and engine failure;

(m) the pre-flight briefing which is given just before each flight, and which consists of a brief summary of the principal parts of the theoretical knowledge course, together with any particular points of airmanship, air traffic control, and meteorology pertaining to the flight; and

(n) the relevant air law aspects.

(4) Practical training course

(a) In-flight instruction

A full briefing must be given during flight, covering the following:

(i) Airmanship –

(aa) The suitability of pick-up and drop areas in respect of size, shape, surface, slope, approach and take-off paths and obstructions;

(bb) Helicopter operation with due regard to such matters as power in the hover, power limitations, hovering into the wind, position of ground crew, and obstructions;

(cc) The limits for the relevant conditions;

(dd) Good lookout at all times;

(ee) Built-up areas and gatherings of people must be avoided when a load is suspended below the helicopter, provided that where the operation is to be conducted within be a built-up area, safe flight routes must be established and approved by the Director and local municipality;

(ff) Cabin crew, if used, must be safely secured to the helicopter at all times by means of a safety harness or seat belt.

(ii) Hook-up and transition –

(aa) Demonstrate the positioning of the helicopter accurately above the load using the techniques of marshalling either by radio, visual signals, mirror or cabin crew intercom;

Note: When a cabin crew member is used for marshalling, the pilot must strictly obey his or her instructions at all times, except if the helicopter and its occupants would be placed in jeopardy by doing so;

(bb) Demonstrate the pick-up and the transition to forward flight when at a safe height;

(cc) The appropriate cruise speed should take into account the load's flight characteristics, the environment, level of turbulence and engine power available;

(dd) Demonstrate control of the load during flight and procedure to be followed if the load becomes difficult or impossible to control. For example, if the load starts oscillating, the pilot should reduce power and enter a gentle turn left or right, or bring the helicopter to a stationary hover; this generally will alleviate the condition. The load should only be jettisoned in extreme cases when the helicopter or its occupants are at risk and then only over uninhabited areas.

(iii) Approach and drop-off –

(aa) The approach should be cautious and fairly shallow, taking into account the distance the load is beneath the aircraft and above the surface;

(bb) The transition to the hover should be made high, to ensure adequate clearance between the load and the surface or ground obstacles;

(cc) Directional information should be provided by the radio, visual signals or cabin crew during the final stages of the approach;

(dd) Demonstrate positioning the load over the drop-off point and lowering it to the surface or its position, using the techniques of marshalling either by radio, visual signals, mirror or cabin crew intercom;

(ee) Demonstrate releasing the load, using the normal release method and the emergency release method.

(iv) Common faults –

(aa) Lack of precision when hovering inside ground effect or outside ground effect;
(bb) Lack of appreciation for ground clearance with an underslung load;
(cc) Vertical drift when lifting and lowering the load;
(dd) Horizontal drift when lifting and lowering the load;
(ee) Jerky pick-up and drop-off;
(ff) Pilot-induced oscillations due to over-controlling on the cyclic;
(gg) The effects of trying to counter oscillations in flight using cyclic instead of power and speed.

(b) Air exercises

Exercise 1: Hook-up procedure

(i) Approach the hook-up area using –
  (aa) ground marshaller;
  (bb) radio;
  (cc) cabin crew intercom; and
  (dd) helicopter mirror.
(ii) Establish a steady hover using –
  (aa) short strop;
  (bb) long strop.
(iii) Once the load has been hooked up, take up the slack while monitoring the power required to hover before lifting the helicopter vertically until the load is well clear of the surface or obstacles, as communicated/established by each of the methods listed under paragraph (a) above.
(iv) Once the load is clear, transit to forward flight.

Exercise 2: In-flight

(v) Observe Vne as established from the flight manual or dictated by the load,while handling the controls as smoothly as possible;
(vi) Reduce power and enter a gentle turn to either left or right, or bring the helicopter to a stationary hover, to demonstrate the technique for bringing an oscillating load under control;
(vii) Avoid any built-up or inhabited areas during flight with a sling load.

Exercise 3: Drop-off procedure

(viii) Approach the drop-off area at a shallow angle using –
  (aa) ground marshaller;
  (bb) radio;
  (cc) cabin crew intercom; and
  (dd) helicopter mirror;
(ix) Terminate the approach in a high hover with the load well clear of the surface or ground obstacles as communicated/established by each of the methods listed under paragraph (a) above;
(x) Maintain a steady inside ground effect hover or outside ground effect hover while monitoring the power required to hover;
(xi) Position the load over the drop-off point;
(xii) Once in position, lower the load vertically until it contacts the surface and then jettison it using –
  (aa) the normal release system; or
  (bb) the emergency release system.

Note: Both normal and emergency release methods are to be practised.

(c) Post-flight discussion

The post-flight discussion reviews the exercise and can be used to amplify or clarify any particular point or difficulty, thus consolidating the exercise as a whole.

(5) Skills test

The applicant shall demonstrate competency in the aspects of subparagraph (b) using appropriate form.

61.23.1 REQUIREMENTS FOR HELICOPTER WINCHING RATING

1. Training

(1) Aim of training course

The aim of the training course is to train a candidate to the level of proficiency required for the issue of a helicopter winching rating, and to provide the training necessary to act as pilot-in-command of a helicopter engaged in winching operations.

(2) Contents and requirements of training course

(a) The candidate must have completed not less than 250 hours of flight time as pilot-in-command of a helicopter. The course must be conducted by the holder of an aviation training organisation approval, issued by the Director in terms of Part 141.

(b) The course comprises –

(i) a theoretical knowledge course; and

(ii) a practical training course.

(3) Theoretical knowledge course

(a) The theoretical knowledge course must comprise instruction on the following –

(i) The significance of operations inside ground effect and outside ground effect and the correct use of the relevant performance charts;

(ii) the preflight checking and correct operation of the helicopter winching equipment, including the emergency cable cutter;

(iii) the marked lateral C of G shift that takes place when winching;

(iv) the importance of a full and correct briefing for all flight and ground crew members participating in the operation as regards to –

(aa) pick-up and drop-off points;

(bb) the care, selection, preparation and correct use of winching equipment, including inspection of cables, nets, strops, and safety harness for winch operator, as applicable;

(v) responsibilities and duties of cabin crew;

(vi) marshalling signals;

(vii) load preparation and flight characteristics of different loads;

(viii) aircraft-generated static, use of the static discharge pole and the correct procedures in this regard;

(ix) correct radio procedures and terminology for intercom communication between the pilot and the winch operator;

(x) pick-up and release procedures;

(xi) emergency procedures, including engine failure in the hover, cables getting fouled either with the helicopter or with other items, loads becoming difficult or impossible to control in flight and cable cutting; the effects of buildings and obstruction on prevailing winds, escape routes in the event of downdrafts, turbulence and engine failure;

(xii) safety and other equipment including hand held transceivers, hard hats, safety goggles, durable gloves, overalls and whistles;

(xiii) the relevant air law aspects;

(xiv) the pre-flight briefing which is given just before each flight, and which consists of a brief summary of the principal parts of the theoretical knowledge

course, together with any particular points of airmanship, air traffic control, and meteorology pertaining to the flight.

(4) Practical training course

(a) In-flight instruction

A full briefing must be given during flight, covering the following:

(i) Airmanship –

(aa) The suitability of pick-up and drop areas in respect of size and shape, surface, slope, approach and take-off paths, and obstructions;

(bb) Helicopter to be operated within its Vne for winching operations at all times. Transition into forward flight should only be undertaken once the winch cable has been safely stowed;

(cc) The limits for the relevant conditions;

(dd) Good lookout at all times;

(ee) Built up areas and gatherings of people must be avoided when a load is suspended below the helicopter, provided that where the operation is to be conducted within a built up area, safe flight routes must be established and approved by Director and the local municipality;

(ff) Personnel must only be raised or lowered by means of the winch from a stationary hover in relation to the surface and never while the helicopter has any apparent forward, sideward or rearward speed unless deemed necessary in the interests of safety;

(gg) The winch operator must be safely secured to the helicopter by a safety harness, particularly when the doors are open during, or have been removed for the operation, and while manoeuvring people into or out of the helicopter cabin;

(hh) The helicopter should never transit to forward flight with people suspended on the winch cable unless deemed necessary in the interests of safety;

(ii) The helicopter must be operated with due regard for such matters as power limitations, hover power, hovering into the wind, and position of obstructions;

(jj) While in the hover during hoisting operations, the pilot must strictly obey the hoist operator’s instructions at all times, except if the helicopter and its occupants would be placed in jeopardy by doing so.

(ii) Hook-up and transition

(aa) Demonstrate the approach and accurate positioning of the helicopter over the pick-up point, guided via the intercom by a flight crew member acting as winch operator;

(bb) Demonstrate the pick-up and the pronounced lateral C of G shift when the winch takes up the weight;

(cc) Demonstrate the transition to forward flight only when the winching operation has been completed. The winching operation is completed when the winched persons or cargo are safely aboard, the winch cable is safely stowed and, where applicable, the cabin doors have been closed.

(iii) Approach and drop-off

(aa) The approach should be normal;
(bb) Control should be handed over to the winch operator when still some distance short of the drop-off point;
(cc) Demonstrate positioning the helicopter, following the instructions given by the winch operator over the intercom;
(dd) Demonstrate the drop-off and the pronounced C of G shift when the load is removed from the winch cable.

(iv) Common faults
(aa) Lack of precision when hovering inside ground effect or outside ground effect;
(bb) Vertical drift when lifting and lowering the load;
(cc) Horizontal drift when lifting and lowering the load;
(dd) Pilot-induced oscillations due to lateral C of G shift and over-controlling on the cyclic when hovering during the pick-up and drop-off.

(b) Air exercises
(i) Exercise 1: Pick-up procedure
(aa) Approach the pick-up area guided by instructions received over the intercom from the winch operator;
(bb) Establish a steady inside ground effect hover or outside ground effect hover over the pick-up point following the winch operator's instructions while he or she lowers the winch cable;
(cc) Indicate the pronounced C of G shift experienced when picking up the load;
(dd) Once the load has been picked up, monitor the power required to hover until the load is safely stowed inside the helicopter, the winch cable is secure and, if applicable, the cabin doors have been closed, before transitioning to forward flight.

(ii) Exercise 2: In-flight
Observe Vne as established from the flight manual.

(iii) Exercise 3: Drop-off procedure
(aa) Approach the drop-off area normally, following instructions by the winch operator given over the intercom;
(bb) Terminate the approach in a high hover and follow the winch operator's instructions for the positioning of the helicopter while he or she is lowering the winch cable;
(cc) Indicate the pronounced C of G shift when the load is released;
(dd) Once the load has been released, maintain the hover until the winch is safely stowed and, if applicable, the cabin doors have been closed;
(ee) Only then transit to forward flight.

(c) Post-flight discussion
The post-flight discussion reviews the exercise and can be used to amplify or clarify any particular point or difficulty, thus consolidating the exercise as a whole.

(5) Skills test
The applicant shall demonstrate competency in the aspects of subparagraph 4.2 using appropriate form.

61.24.1 REQUIREMENTS FOR HELICOPTER GAME OR LIVESTOCK CULL RATING

1. Training

(1) Aim of training course

The aim of the training course is to train a candidate to the level of proficiency required for the issue of an helicopter game or livestock cull rating, and to provide the training necessary to act as pilot-in-command of a helicopter engaged in game or livestock cull operations.

(2) Contents and requirements of training course

(a) The candidate must have completed not less than 250 hours of flight time as pilot-in-command of a helicopter. The course must be conducted by the holder of an aviation training organisation approval, issued by the Director in terms of Part 141.

(b) The course comprises –

(i) theoretical knowledge course; and

(ii) practical training course.

(3) Theoretical knowledge course

(a) The theoretical knowledge course must comprise instruction on the following:

[Under development]

(4) Practical training course

(a) In-flight instruction

[Under development]

(5) Skills test

The applicant shall demonstrate competency in the aspects of paragraph 4 using Appropriate form.

61.25.2 SKILLS TEST FOR AGRICULTURAL PILOT RATING

1. Conducting the Skills Test

The person conducting the skills test shall test an applicant for the issuing of an agricultural pilot rating on his or her ability to perform as pilot-in-command of an aeroplane, helicopter or microlight aeroplane, as the case may be, in the following procedures and manoeuvres with a degree of competency appropriate to the privileges granted to the holder of an agricultural pilot rating:

(1) The skills test shall be conducted in accordance with the skills test appropriate form and shall include the following –

(a) Assessment of area to be sprayed

(b) Load sheet

(c) Weather report

(2) In the case of aeroplanes –

(a) Short-field take-off and landings

(b) Cross-wind and down-wind take-offs and landings;

(c) Flight manoeuvres at minimum air speed;

(d) Accelerated stalls;

(e) Maximum-rate turns;

(i) Incipient spin recoveries entered into inside of and from outside of turns;

(ii) Precision landings, normal, down-wind and cross-wind;

(iii) Exit from application area, turn around and re-entry to application area under various wind conditions;

(iv) Simulated application runs at appropriate heights;

(v) Entry to and exit from applications over obstructions;

(aa) Avoidance of obstructions;

(bb) Emergency procedures.

(vi) Low-level forced landing technique

(vii) Dump load

(3) In the case of helicopters –

(a) Take-offs and landings at maximum certificated mass for aerial applications;

(b) Cross-wind and down-wind take-offs and landings;

(c) Flight manoeuvres at minimum air speed;
(d) Maximum-rate turns;
(i) recoveries entered into inside of and from outside of turns;
(ii) Precision landings, normal, down-wind and cross-wind;
(iii) Exit from application area, turn around and re-entry to application area under various wind conditions;
(iv) Simulated application runs at appropriate heights;
(v) Entry to and exit from applications over obstructions;
(aa) Avoidance of obstructions;
(bb) Emergency procedures;
(vi) Low-level autorotation technique;
(vii) Dump load.

2. Skills Test Standard

The skills test shall be conducted in accordance with the standard contained in the skills test appropriate form.

61.25.3 APPLICATION FOR AGRICULTURAL PILOT RATING

1. Skills test report

(1) The skills test form, completed by the Designated Flight Examiner, shall accompany the application form.
(2) The Agricultural Pilot Rating shall be endorsed in the pilot licence.

61.26.2 GENERAL REQUIREMENTS FOR DESIGNATED FLIGHT EXAMINERS

1. Assessment course

A Designated Flight Examiner is required to successfully complete the flight examiner assessment course, the contents of which are stated below.

(1) Flight examiner assessment course
(a) An applicant for his or her first designation as a flight examiner (DFE) or for approval as a person to act as flight examiner (APFE) shall have completed an approved flight examiner assessment course within the 12 months immediately preceding such application;
(b) The assessment course, referred to in paragraph (a), shall be conducted by the CAA or by the holder of an aviation training organisation approval issued in terms of Part 141, and shall include – as appropriate to the role of the examiner – at least the following subjects –
(i) the regulatory requirements relevant to the examination duties;
(ii) fundamentals of human performance and limitations relevant to flight examination;
(iii) fundamentals of evaluation relevant to examiner's performance;
(iv) CAR Part 61, related technical standards, AIP, published flight guides, and AICs;
(v) quality system as related to CAR Part 61;
(vi) multi-crew co-operation, and human performances and limitations, if applicable;
(vii) constructing and conducting a skills test or proficiency check; and
(viii) performance assessment.

61.26.10 CONDUCTING OF SKILLS TEST AND PROFICIENCY CHECK BY DESIGNATED FLIGHT EXAMINERS

1. Guidelines

The guidelines for the conducting of a Skills Test or Proficiency Check are contained in Appendix 9.0 to SA-CATS 61.